University of Nottingham
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Department of Mechanical, Materials and Manufacturing Engineering

Investigation on surface integrity of Ti-6Al-4V alloy after laser assisted machining

BEng PROJECT FINAL DISSERTATION 2023-24

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Date of report: 2^{nd} May 2024

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Summary

This report details an experimental based bachelor's individual project that investigated the effects of laser-assisted machining (LAM) on the surface integrity and functional changes of Ti-6Al-4V (Ti64) alloys by experiment and data analysis in order to understand the deformation mechanism and chemical reaction in the cutting process. Currently, Ti64 alloys are widely used in high-end technology industries such as aerospace, thanks to their excellent mechanical and chemical properties like high strength, corrosion resistance and high temperature resistance. However, its superior properties make the machining of titanium alloys a challenge. Conventional machining (CM) methods can compromise the surface integrity of titanium alloys and thus affect their functionality.

Therefore, this project is dedicated to exploring the impact of laser-assisted milling (LAMill), an innovative thermally assisted machining (TAM) method that applies laser to soften the cutting material, on the surface integrity of the machined Ti64 alloys, including the cutting force and tool wear in cutting process; the surface results of the workpiece and the chips and the subsurface results and the chemical contents of the machined workpiece. This research could lead to optimised machining methods and improved product quality for certain key components in high-end manufacturing.

In this project, three types of machining method were conducted to obtain an accurate result, including Conventional Milling (CMill), LAMill and Laser scan (LS) to the Ti64 alloy. The experiment of the LAMill to the Ti64 alloy was conducted based on a novel thermal placement control method which could achieve the uniformly heating of the whole workpiece. Three different temperatures were selected for LAMill and LS based on the internal phase transition of the titanium alloy, which are 650 degrees Celsius (650°C), 850 °C and 1050 °C. The processed workpieces were used to analysis the results of the surface and then were cut and sampled to analysis the results of the subsurface. It was found that LAMill reduced the cutting forces and tool wear and produces flatter chips than CMill of Ti64 alloys. In addition, the surface roughness of the workpiece was reduced with LAMill and the subsurface deformation was also reduced. Also, it was found that introducing lasers into the milling process did not cause the same destructive results as LS alone. All results suggest that LAMill of Ti64 alloys can actually improve surface integrity after machining compared to the CMill. And of the three temperatures selected, it was found that LAMill at 650 °C was not as well optimised for surface integrity as at 850 °C and 1050 °C and that 1050 °C resulted in excessive material removal. Therefore, 850 °C is expected to be the optimum parameter for LAMill of Ti64 alloys.

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Contents

NOMENCLATURE

Symbol	Quantity	Unit
Fx	Cutting force in x direction	N
Fy	Cutting force in y direction	N
Fz	Cutting force in z direction	N
Fa	Combined cutting force	N
Ra	Surface roughness	µm
Rt	max value of surface peak to valley	µm
°C	degree Celsius	Degrees Celsius

ACRONYMS

Acronym	Meaning
CM	Conventional machining
CMill	Conventional milling
CFS	Chip front surface
CTES	Chip top edge surface
CBES	Chip bottom edge surface
EDX	Energy dispersive X-ray
EDS	Energy dispersive X-ray Spectroscopy
LAM	Laser-assisted machining
LAMill	Laser-assisted milling
LS	Laser scanning
RT	Room temperature
SEM	Scanning electron micro-scape
Ti64	Ti-6Al-4V alloy
TAM	Thermal assisted machining

1. INTRODUCTION

1.1. Background and context

In recent years, Ti-6Al-4V (Ti64) alloys are widely applied in aerospace, nuclear and biomedical industries due to their excellent mechanical properties and chemical inertness, such as high strength, corrosion resistance and high temperature resistance [1]. However, titanium alloys have very poor process properties. Machining of Ti64 alloys is one of the most difficult industrial challenges faced as their low thermal conductivity, low elastic modulus, and high chemical reactivity. These properties can lead to high tool wear and poor surface integrity easily with conventional machining (CM) method [2]. Meanwhile, final surface integrity is often a determining factor in product quality and performance. Specifically, surface integrity is one of the key criteria for determining whether safety-critical components in some industry's meet the stringent standards for functional performance. Therefore, to improve the machinability and hence the product quality of these difficult-to-cut materials, thermally assisted machining (TAM) methods have been proposed, whereby the workpiece is locally preheated before the tool, thus softening the material at high temperatures [3].

Laser-assisted machining (LAM) is one of the most widely used thermally assisted machining (TAM) methods today. LAM is an innovative method that combines high-power laser localised heating of the workpiece with material removal by conventional tools as shown in figure 1 [3]. This method can decrease the cutting force and increase tool life compared to CM methods [3]-[4]. Currently, LAM is mainly used for turning or milling operations, where a single laser spot is usually placed in front of the tool as a heat source to validate the effectiveness of the LAM approach. Although the mismatch between the laser spot size and the cutting width, resulting in uneven heating of the workpiece, once an obstacle to the further application of laser-assisted processing, a recently proposed thermal placement control method that simultaneously considers both forward and inverse issues in the laser-assisted milling (LAMill) process solves this problem well. This method utilises a small spot size to achieve a uniformly heated cut around the edges of large workpieces and significantly improves material removal rates. [3]

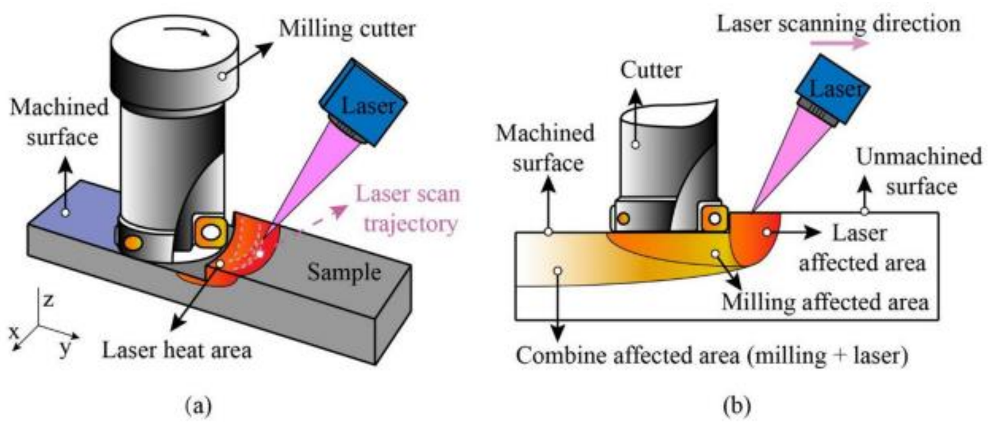

Figure1. Schematic and equipment of LAM: (a) Laser scanning trajectory for evenly 3D heating the workpiece and (b) heat influenced superficial layer [3].

Although LAM has been shown can Improve processability of certain materials and LAM technology is constantly being improved, the impact of this technology on high performance alloys such as titanium alloys still needs to be investigated in-depth, especially the changes in surface integrity and functionality due to high thermal energy. In addition, residual stresses on the workpiece caused by thermal and mechanical energy during machining also need to be studied in- depth as the residual stresses generated in the machining of titanium alloys have a significant impact on product safety [5]. However, there is not enough research to prove that this machining approach meets our requirements. Therefore, this project is dedicated to the study of the effects of LAM on the surface integrity changes (including detailed microstructure changes and chemical reactions in machining processes) of Ti64 alloys.

1.2. Aim and Objectives

The aim of this project is to obtain the effect of LAM (Specifically referring to LAMill) on the surface integrity of Ti64 alloy and the underlaying mechanism by observing the surface and subsurface analysis results on the machined workpieces and cutting chips. The project will also investigate functional changes in Ti64 alloys under laser irradiation. The research can be used to optimise the machining and improve the quality of certain critical aerospace component structures.

The objectives are as the following points:

• Identify forces and tool wears on Ti64 alloys under the influence of LAM and analyse the data to compare the impact of LAM processing with traditional machining methods.

• Identify and analyse the surface results of workpiece and cutting chips to obtain the surface integrity on the surface and the change of material removal mechanism.

• Identify and analyse the subsurface results including microstructure, microhardness and other chemical contents to have in-depth understanding of the surface integrity.

• Recommend optimised parameters in LAMill Ti64 alloys to improve the machinability and obtain the surface integrity improvement

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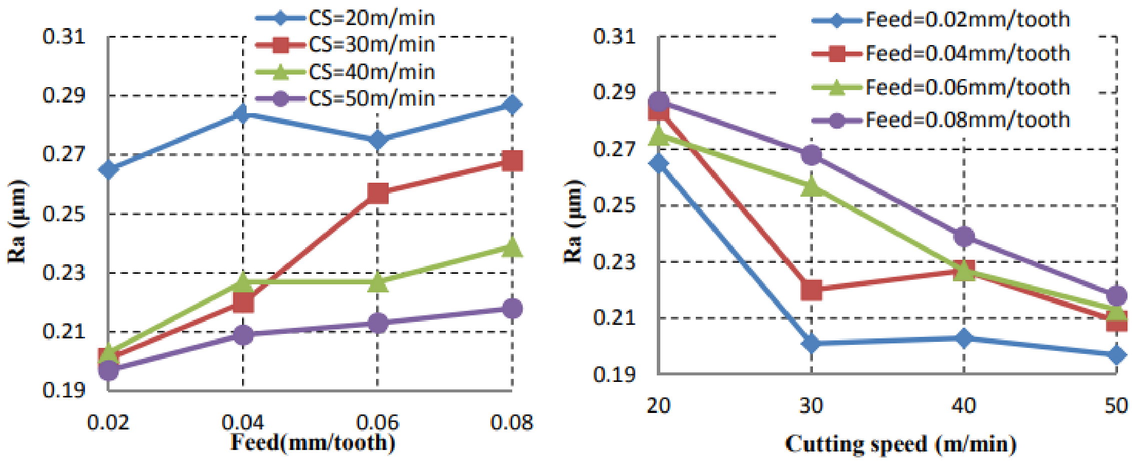
2. LITERATURE REVIEW

In recent years, Ti64 alloys have attracted widespread attention in high-end manufacturing industries such as aerospace, nuclear and medical device manufacturing due to their excellent material properties, such as excellent strength-to-weight ratio and strong corrosion resistance. However, the application rate of titanium alloys in these fields has not increased significantly due to their difficult machining characteristics [4]. Therefore, many researchers have initiated studies on the machining methods of titanium alloys. After reviewing the existing literature, it is found that the conventional milling (CMill) method of machining Ti64 alloys has been well researched and this method has been proved to be inefficient. LAMill has also been extensively studied as a more efficient thermally assisted machining. However, there are not many detailed studies on the surface integrity of titanium alloys subjected to machining processes.

2.1 Conventional machining of Titanium alloy

In CMill, the surface roughness (Ra) of titanium alloys is affected by machine speed, milling cutter type, feed rate and depth of cut [6]. Since surface roughness is related to the performance of the material, the study of these machining parameters is necessary. A full factorial experiment was conducted to investigate the effect of cutting speed as well as feed rate on the surface roughness of titanium alloys during milling. The results in figure 1 show that the surface roughness increases slightly with increasing feed rate and decreases with increasing cutting speed [7].



(a) How feed rate effect the Ra (b) How cutting speed effect the Ra

Figure 1. Effect of a) feed and b) cutting speed to the surface roughness [7].

In addition, the effect of feed and cutting speed on the cutting force during machining was investigated and the results in figure 2 shows that the cutting force generally increases with the increase in feed and decreases with the increase in cutting speed [8].

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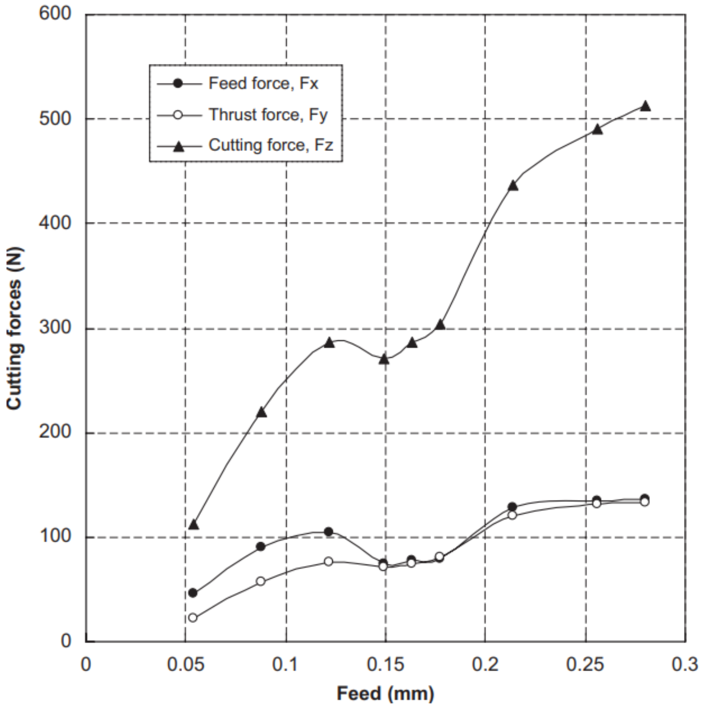

Figure2. Cutting force versus Feed [8].

For traditional milling methods, due to the low thermal conductivity of titanium alloys and the extreme tendency to alloy, the problem of tool failure due to excessive friction and heat in the cutting zone is common during machining [4]. Therefore, the most widely studied problem in CMill for machining titanium alloys is tool wear There are about four types of wear which are abrasion, chipping, adhesion, and plastic failure.

Abrasive wear is produced by friction between the tool and the surface of the workpiece during machining. In the milling of titanium, high cutting forces and high friction often lead to severe abrasive wear and this type of wear can be reduced by apply showed that lubrication at the tool-workpiece interface. In addition, cooling the temperature of the machining area can reduce diffusive wear [9]. Chipping wear is caused by chips shed from the cutting edge and tip during the cutting process and as the cutting speed increases, the cutting percentage gradually increases [8], it can be minimised with the MQL machining condition (shown in figure 3) [9]. In addition, machining in the MQL condition has been shown to have the least amount of tool surface adhesion. In the case of plastic failures, although the probability of occurrence is low, the consequences are often catastrophic and there are no good solutions for such failures. This is also one of the reasons why LAMill of titanium alloys needs to be explored.

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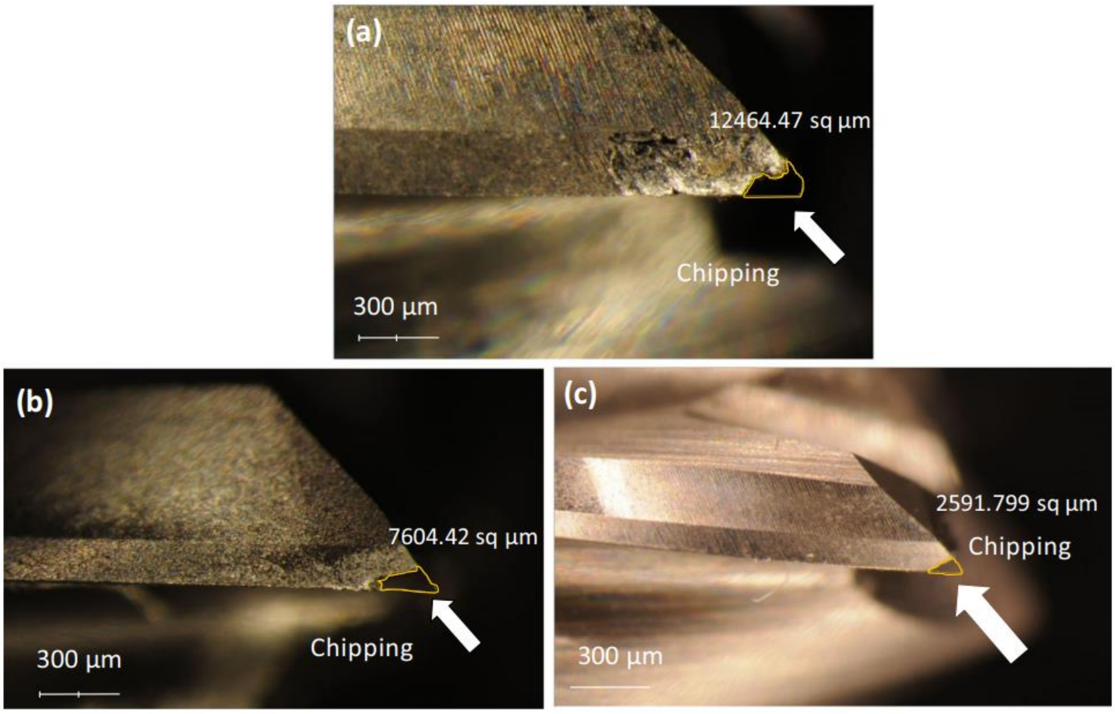
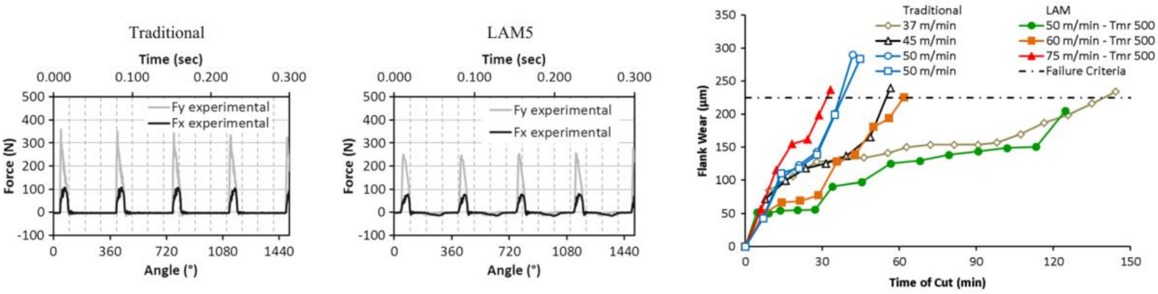


Figure 3. Comparison of chipping wear under a) dry, b) MQL, c) conditions at a constant setting of parameters [9].

2.2 Laser assisted machining of Titanium alloy

LAM is one of the most widely used TAM methods, which could reduce the yield stress by locally softening the machined workpiece leading to the deduction of machining forces and the prolonged tool life [3]. A considerable amount of research has already shown that LAM of titanium alloys have advantages over CM, both in terms of the surface integrity of the machined surface and the effect on the tool.

One study compared CM and LAM of Ti64 alloys at different cutting speeds and the results shown in figure 4 revealed that LAM of titanium alloys reduced cutting forces during machining and effectively improved tool flank wear [10].



(a)cutting force with CM (b)cutting force with LAM (c)flank wear with CM and LAM

Figure 4. Comparison of a) b) cutting force and c) flank wear of CM and LAM [10].

In addition, after comparing the residual stresses in the machined workpieces, the results in figure 5 show that LAMill could reduce the residual stresses in the workpieces to a certain extent due to the reduce of the cutting force which leading to a slight decrease in surface hardness [10].

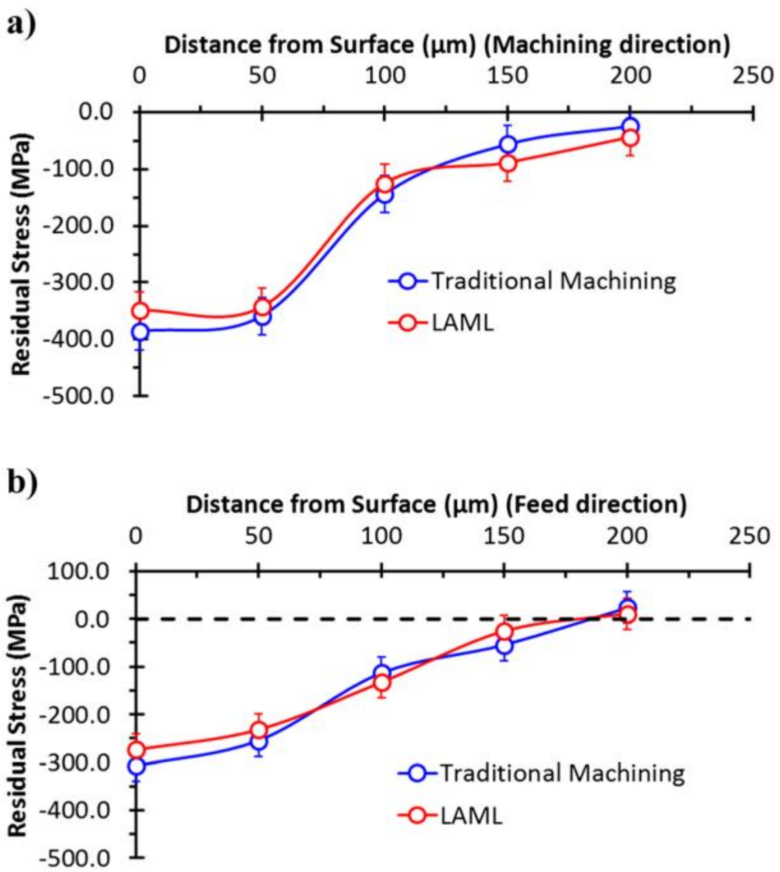


Figure 5. Comparison of residual stresses of CMill and LAMill in different a) Machining and b) feed directions [10].

Kalantari et al. also confirmed that the surface hardness after LAM machining is slightly less than that of CM after comparing the microhardness changes between LAM and CM at different feed rates (shown in figure 6), which could be caused by plastic deformation induced at higher cutting depths [11].

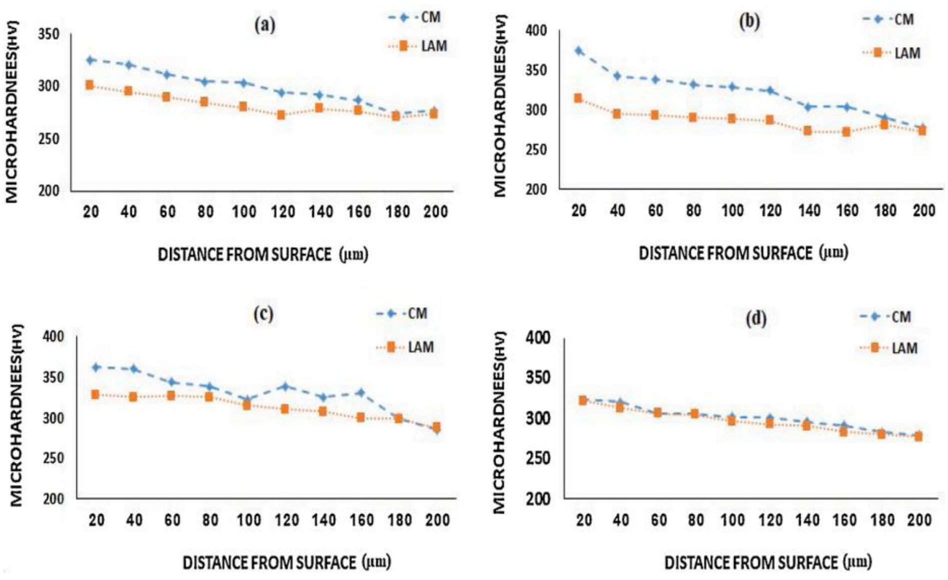


Figure 6. Microhardness variation in LAM and CM process at feed rate of: (a) 0.065 mm/rev (b) 0.104 mm/rev (c) 0.174 mm/rev (d) 0.261 mm/rev [11].

Surface roughness is one of the most important parameters in terms of its influence on the surface integrity of the material. In LAM, a better surface integrity should be expected as the softening of the workpiece attenuates vibrations during machining, and it is confirmed by the results of Kalantari et al' s (figure 7) -the surface roughness after LAM machining is generally lower than that after CM machining at different cutting speeds, depths of cut and feed rates which may be caused by the thermal effect in LAM [11].

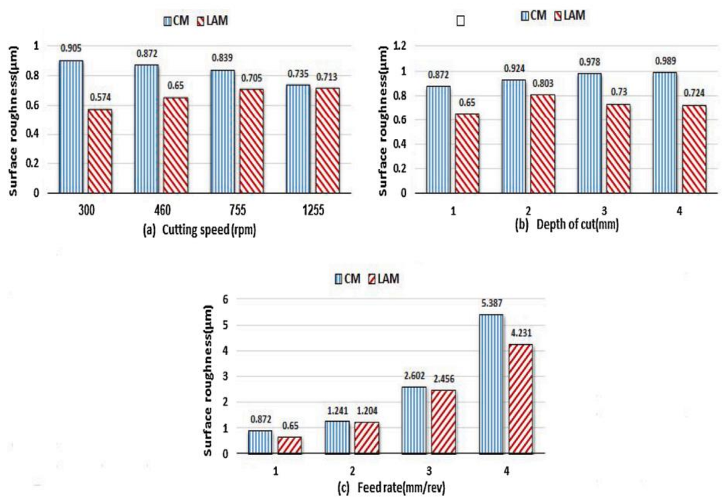


Figure 7. The surface roughness value after LAM and CM process at the different: (a) cutting speed, (b) depth of cut, and (c) feed rate [11].

The thermal effect during the process of laser heating of the workpiece would affects the internal grain changes in the material. LAMill of titanium alloys produced larger grain sizes under all conditions tested (as observed in figure8) [11].

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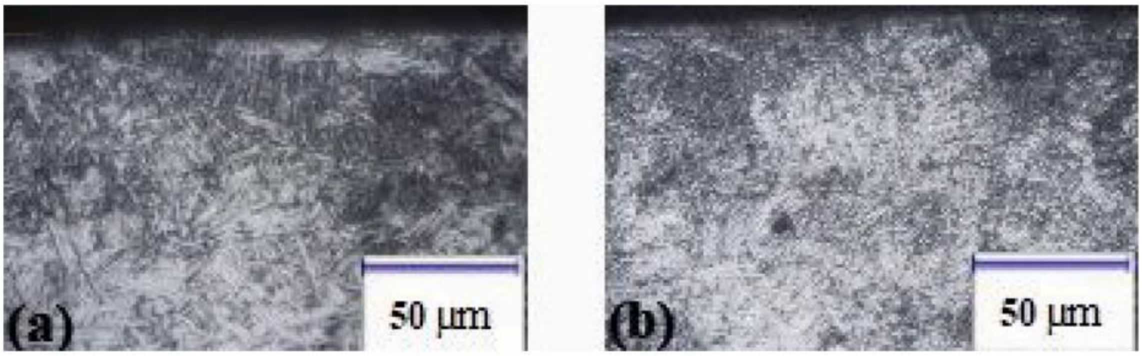


Figure 8. Grain size comparison after a) LAMill and b) CMill [11].

This thermal effect should be avoided as much as possible during testing, excessive temperatures can also change the microstructure of titanium alloys. When the temperature reaches 700 °C, the titanium alloy precipitates a granular beta phase, which becomes massive when it reaches 900 °C. After 900 °C, the beta phase gradually disappears which could affect the properties of the material [12].

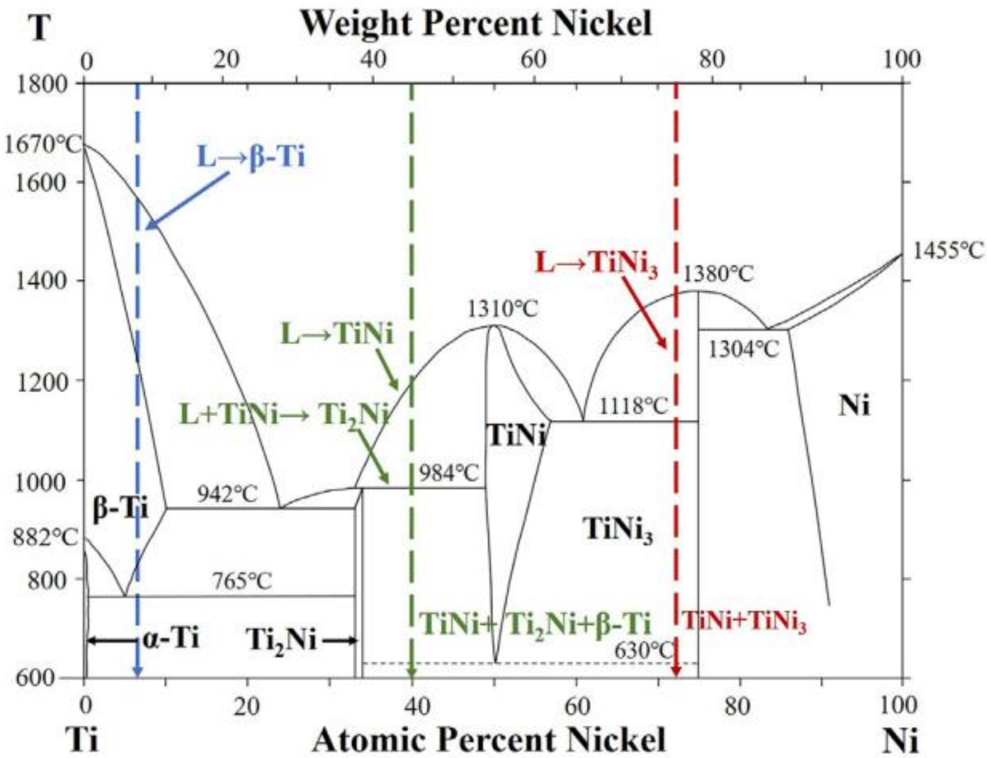


Figure 9. Phase transition of Ti-Ni alloy [12].

In general, there are relatively perfect studies on tool wear during LAM of titanium alloys but further tests are needed for the microstructural changes, surface roughness, microhardness, etc to investigate the surface integrity.

3. METHODOLOGY

3.1 Processing of Ti64 alloys workpieces

The experiment of machining the Ti64 alloy workpiece was conducted based on a test plan which has been shown in 3.1.1. The instruments used in the experiment and the operations are presented in section 3.1.2.

3.1.1 Test plan

The test was divided into three parts, which are Laser-assisted milling (LAMill), pure laser scanning (LS) and conventional milling (CMill) of Ti64 alloys to investigate the effect of the surface integrity of Ti64 alloys under different processing methods and identify the effect of the introduction of lasers in the milling process

The dimension of the machined workpiece and the chosen parameters of the machining process are shown in table 1 and table 2 below respectively.

	Length (mm)	Width (mm)	Height (mm)
Workpiece dimension	201	11.4	41

Table1. Dimension of the processed Ti64 alloy workpiece.

Test description	Laser-assisted milling (LAMill)	Laser scanning (LS)	Conventional milling (CMill)
Laser power (w)	650,850,1050	650,850,1050	-
Laser Scanning speed (mm/s)	3.07	3.07	-
Laser Scanning length (mm)	40	40	-
Cutting speed (m/min)	50	-	50
Cutting depth (mm)	0.5	-	0.5
Cutting width(mm)	11.4	-	11.4
Feed rate (mm/tooth)	0.1	-	0.1

Temperature (degrees Celsius)	RT,650,850,1050	650,850,1050	-

Table 2. Test parameters.

The justifications for the chosen parameters are shown in table 3 below.

Test description	Justification
Laser power (w)	A test set was performed to determine the laser power for heating to different temperatures required in the same amount of time. After testing, it was found that the workpiece could be heated to the desired temperature in the same amount of time using the same laser power as the desired temperature. This is why laser power and temperature are the same.
Cutting speed (m/min)	The official recommended cutting speed for titanium alloy of the used cutting tool is 55m/min and it is known that with the increase of laser power the cutting force will reduce about 10% when the cutting speed is between 25-125 m/min. There would be a shorter tool life if the speed is over 150 m/min [1]. Thus, 50 m/min was chosen to be compared.
Cutting depth (mm)	According to the recommended parameters of the cutting instrument.
Cutting width(mm)	According to the dimension of the workpiece.
Feed rate (mm/tooth)	According to the recommended parameters of the cutting instrument.

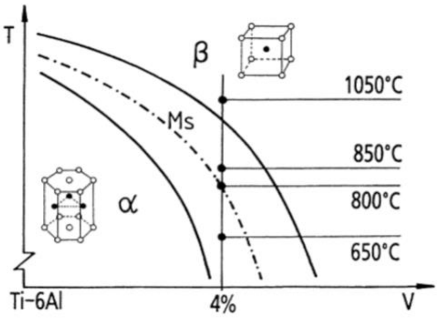
Temperature (Degrees Celsius)	Ti64 alloys have two phases with different properties, alpha and beta, and vary with temperature. Changes in material properties resulting from internal phase transformations in titanium alloys can affect the machining process, so the temperature was chosen based on the phase transformation process within the ti64 alloy as shown in figure 10 [13].  Figure 10. Phase transition of Ti64 alloy with increasing temperature [13].

Table 3. Justifications for selected parameters.

Laser trailing tests:

Three groups of tests were conducted for the trail of pure LS of Ti64 alloy and the parameters are shown in table 4 below.

No	Feed speed(mm/min)	feed speed(mm/s)	Temperature at °C
1	184	3.07	650
2	184	3.07	850
3	184	3.07	1050

Table 4. Parameters for laser trailing tests.

Milling tests:

For the CMill and LAMill of the Ti64 alloys, experiments were carried out at four different temperatures using tools made of Carbide. The detailed experimental parameter settings are shown in Table 5.

No	Temperature	Insert	V(m/min)	V(rpm)	f(mm/tooth)	f(mm/min)	f(mm/s)
1	RT	Carbide4-6, edge1	50	612	0.1	184	3.07
2	650 °C	Carbide4-6, edge2	50	612	0.1	184	3.07
3	850 °C	Carbide4-6, edge3	50	612	0.1	184	3.07

4	1050 °C	Carbide4-6, edge4	50	612	0.1	184	3.07

Table 5. Parameters for CMill and LAMill tests.

3.1.2 Experimental methodology

The uniform heating of the workpiece surface used to be a challenge in LAM processing. In this study, the LAMill experiment was conducted based on a novel inverse control method, which achieved a homogeneously heating of the whole cutting surface of the workpiece with the use of optimised laser trajectories under different laser powers (650w, 850w, 1050w). A comparison of the novel control method and the conventional method has been shown in figure 11 [14].

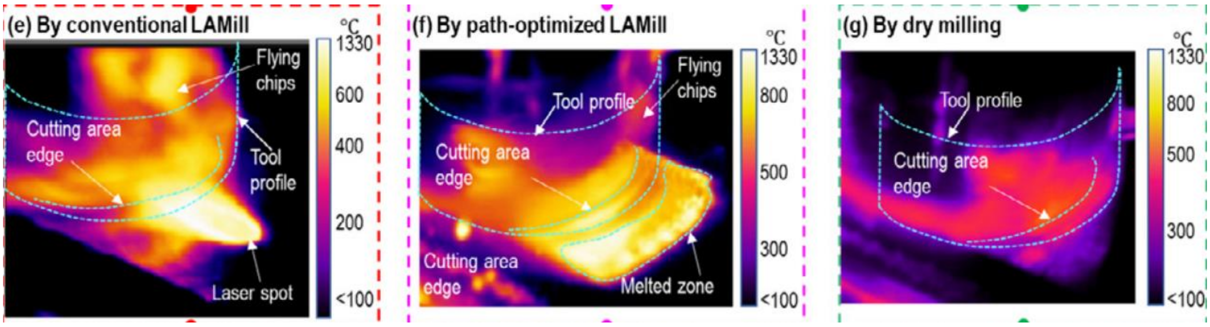


Figure 11. Comparison of temperature distribution of different method [14].

In this study, a powerful and high-efficiency CNC milling machine (Tormach 1100MX) with a 10kw laser system and the cutting tool SEEX09T3AFN-M05 were used for the tests of laser trailing, CMill and LAMill. A FLIR A655SC Long Wavelength IR thermal camera was placed to record the temperature distribution during the LS and LAMill process to make sure the needed temperature has been achieved. Details of the instruments are shown in figure 12 below.

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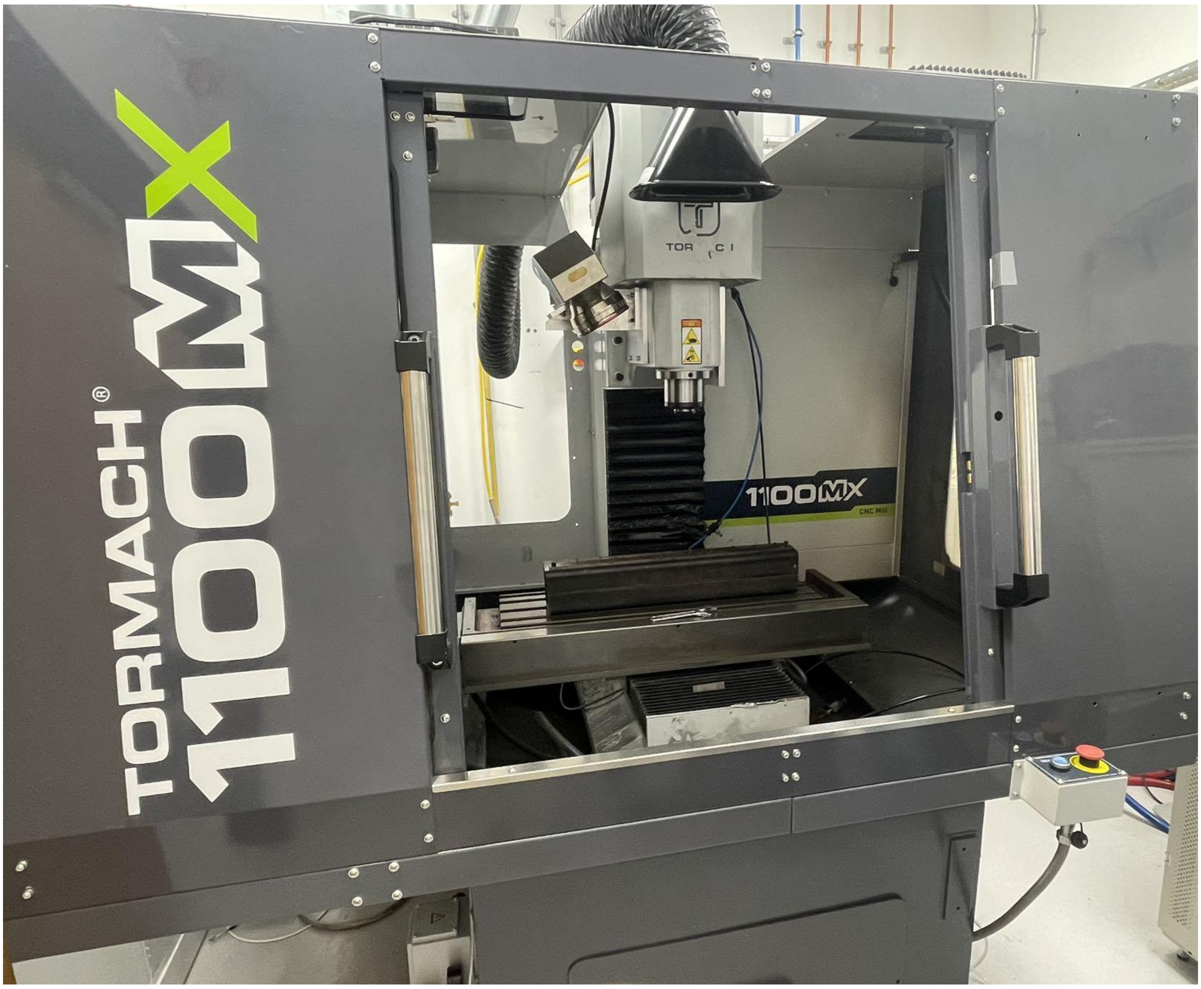
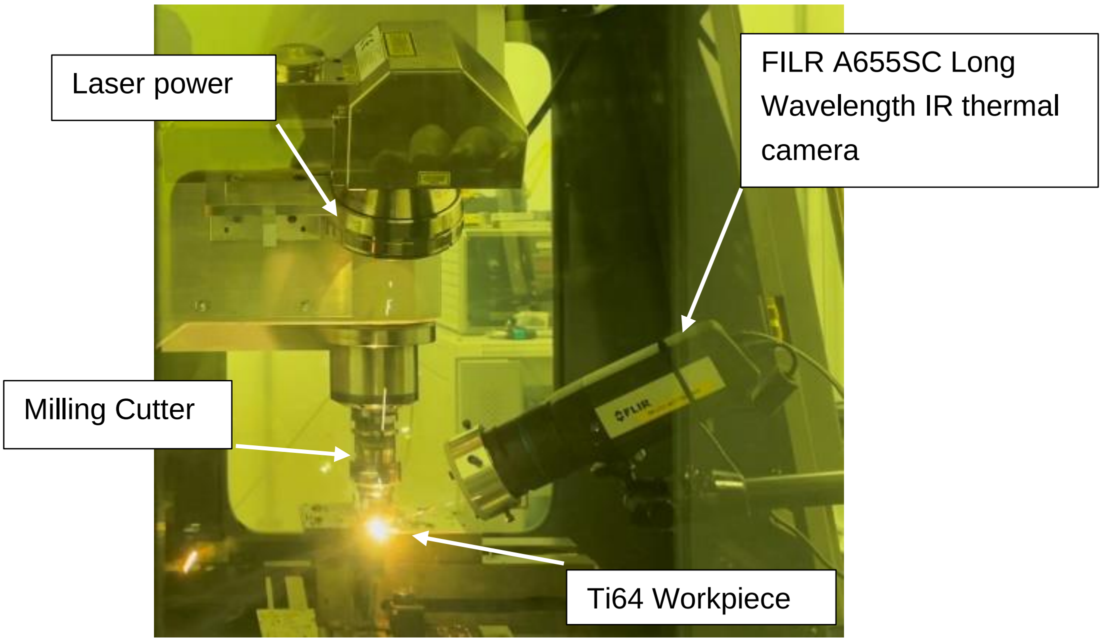

TORMACH®
1100MX
1100MX

(a)


Laser power
FILR A655SC Long Wavelength IR thermal camera
Milling Cutter
Ti64 Workpiece

(b)

[redacted]

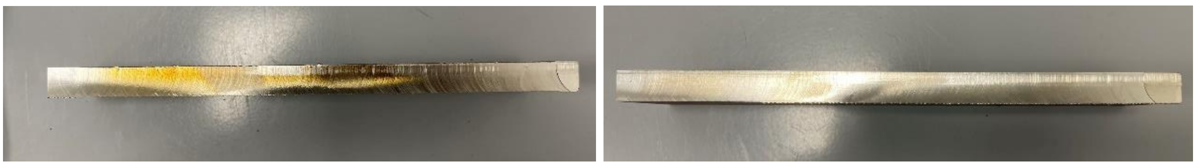
(c)

Figure 12. (a) Tormach 1100MX; (b) Detailed configuration; (c) SEEX09T3AFN-M05 cutting tools.

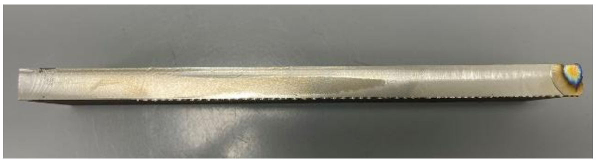
In addition, the Kistler 9257A dynamometer (as shown in figure 13) was used during the CMill and LAMill tests to record the cutting force for analysing.

Figure 13. Kistler 9257A dynamometer.

The machined workpieces are shown in figure 14 and after the milling process in CMill and LAMill, the cutting chips were collected for further analysis (as shown in figure 15).



(a) (b)



(C)

(d)

(e)

Figure 14. Machined Ti64 workpiece under a)1050 DOC LAMill; b) 850 DOC LAMill; c) 650 DOC LAMill; d) RT CMill; e) pure LS.

Figure 15. Collected cutting chips.

3.2 Analysing of the effect to the surface integrity

To investigate the effect to the surface integrity of Ti64 alloys with different processing method, especially LAMill, three aspects were analysed, including the effect on cutting process, the effect on machined Ti64 surface and the effect on machined Ti64 subsurface.

It is worth noting that only the basic research methodology is presented in this section, while the reasons about why the research factors were chosen will be presented in the results section

[redacted]

3.2.1 The effect on Cutting process

The results of cutting force, cutting chips and tools wear were collected and analysed to investigate the effect on cutting process with different processing method.

The cutting force which collected by the Kistler 9257A dynamometer were initially read and parsed via the NI DAQexpress software, which is an interactive software that can view, record, and explore the data. These data were then transformed into matlab for further analysis and comparison. Detailed code on parsing the data is provided in the appendix.

To investigate the effect to the cutting chips and tools wear between different machining process, scanning electron micro-scape (SEM: XL30) (as shown in figure 16) was used to check the microstructure of the chips and tools, including the deformation mechanism, etc. Detailed results will be given in the results and discussion section.

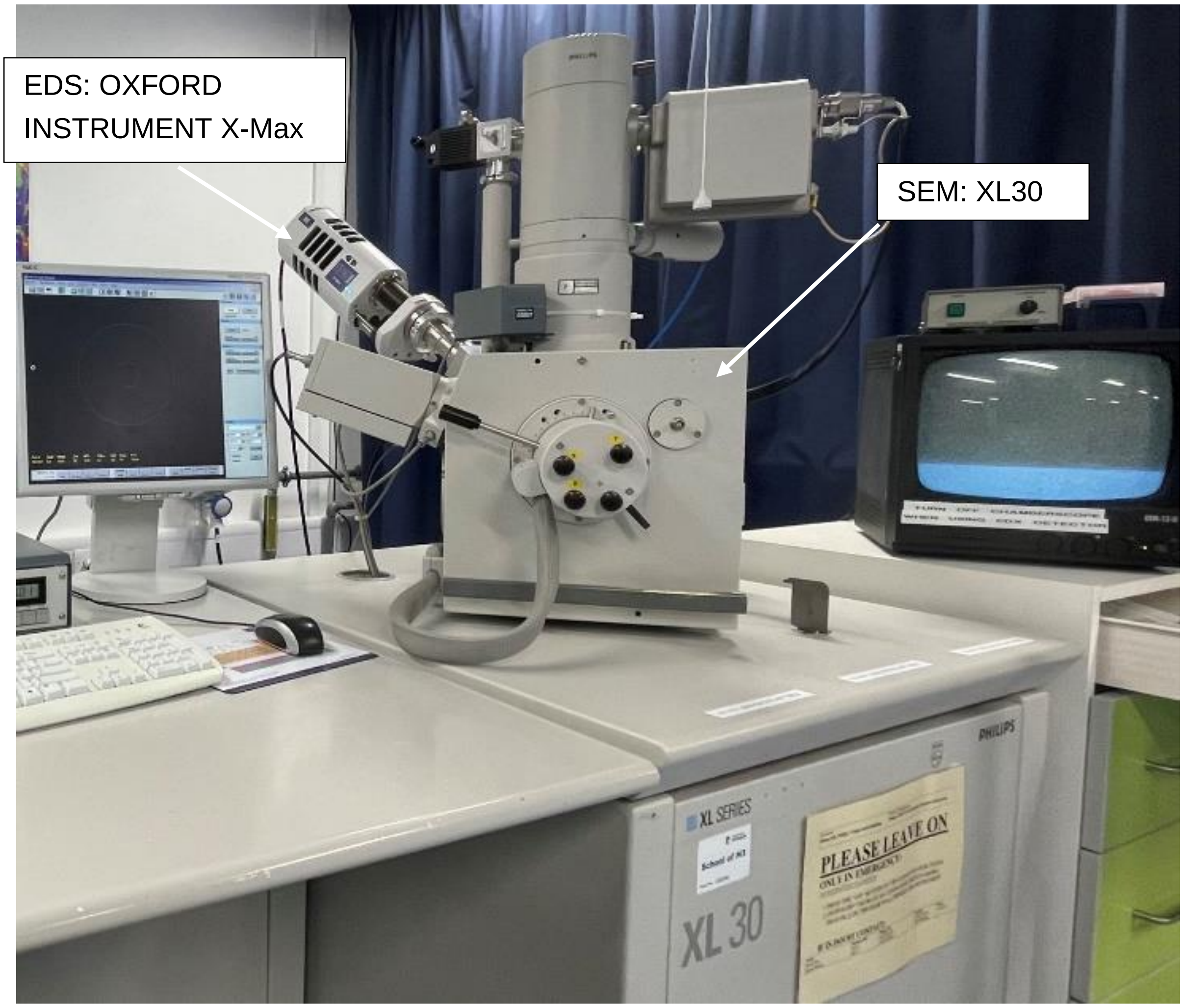


Figure 16. SEM and EDS instrument.

3.2.2 The effect on Ti64 surface

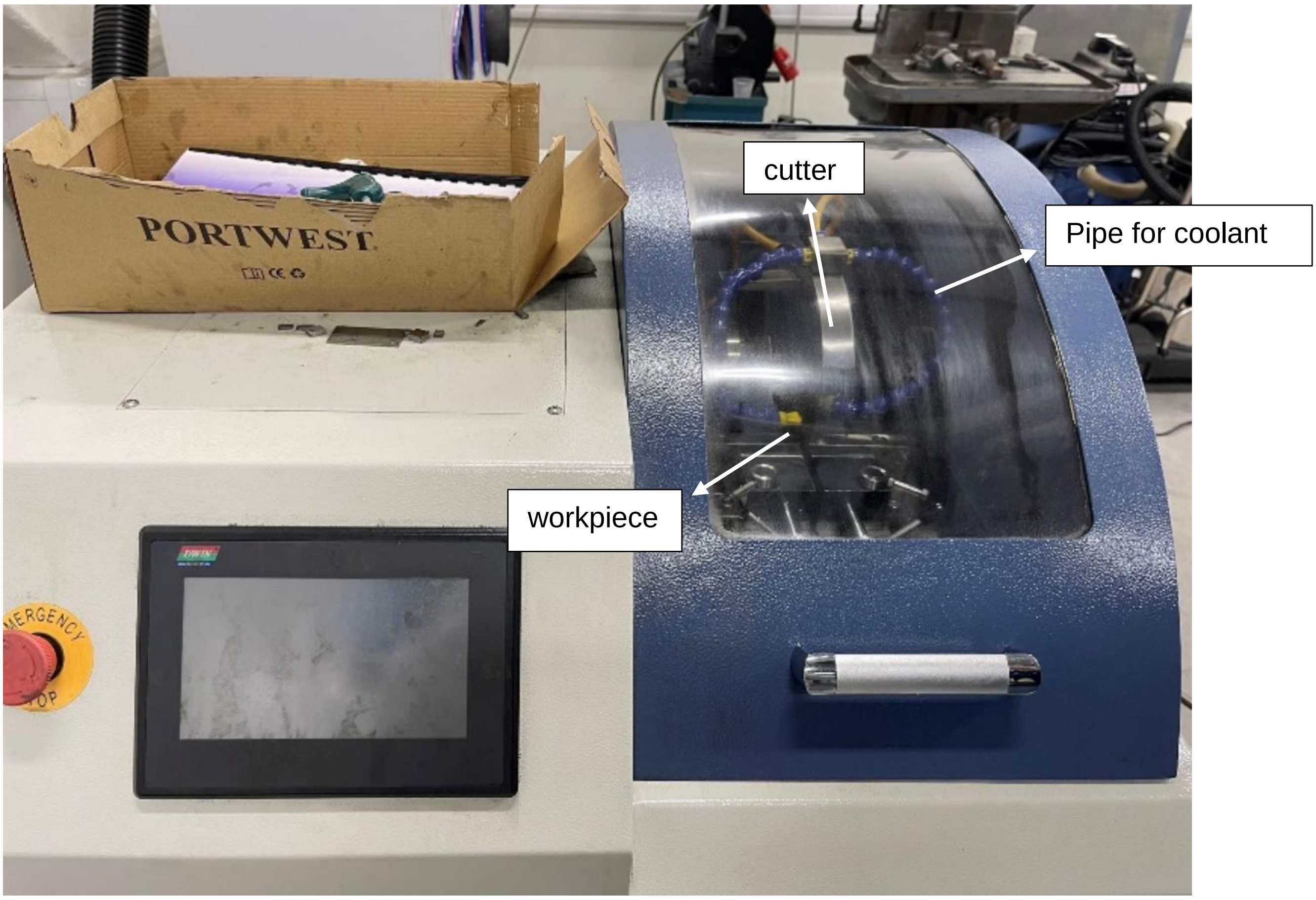
To investigate the effect on the surface, the machined workpiece has been analysed by using surface topography measurement with Alicona G4 Infinite Focus (as shown in figure 17). The results were post-processed using excel and profilm.

Figure 17. Alicona G4 Infinite Focus.

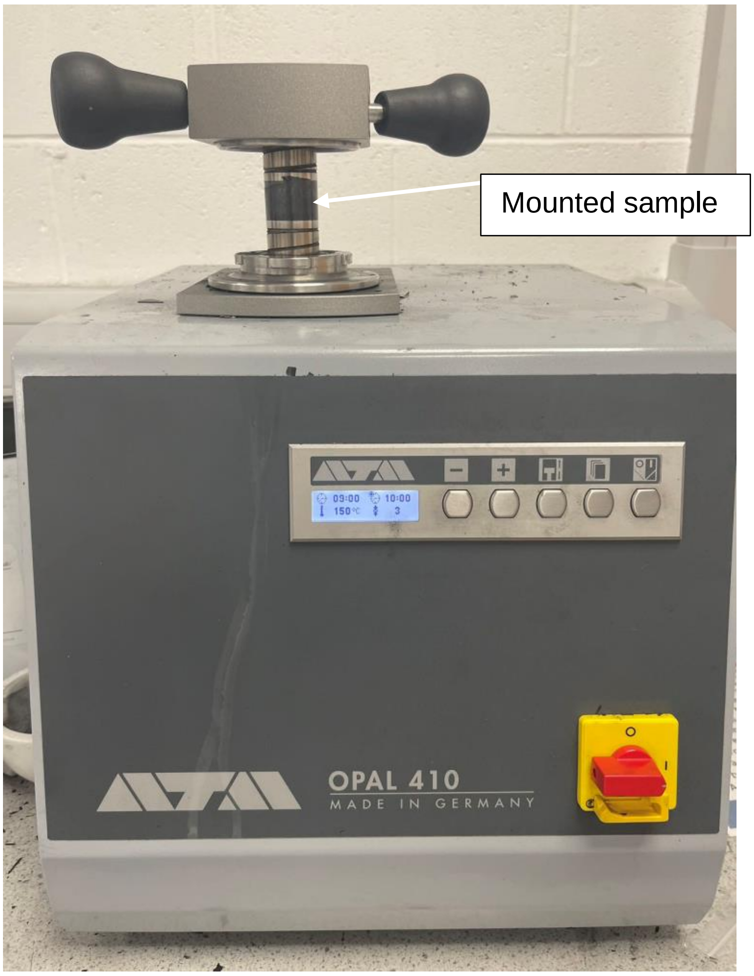
3.2.3 The effect on Ti64 subsurface

In order to investigate the effect of different machining methods on the subsurface of Ti64 alloys, machined workpieces were fabricated as samples for observation. The sample making procedure is as follows.

Firstly, the processed Ti64 workpieces were cut into small squares by the JMQ-60Z cutting machine (as shown in figure 18 a)) in preparation for sampling. Then the cut squares were placed into OPAL 410 (as shown in figure 18 b)) for hot mounting. The scheme for the cutting and mounting is shown in figure 19.



(a)



(b)

Figure 18. a) JMQ-60Z cutting machine; b) Hot mounting press OPAL 410.

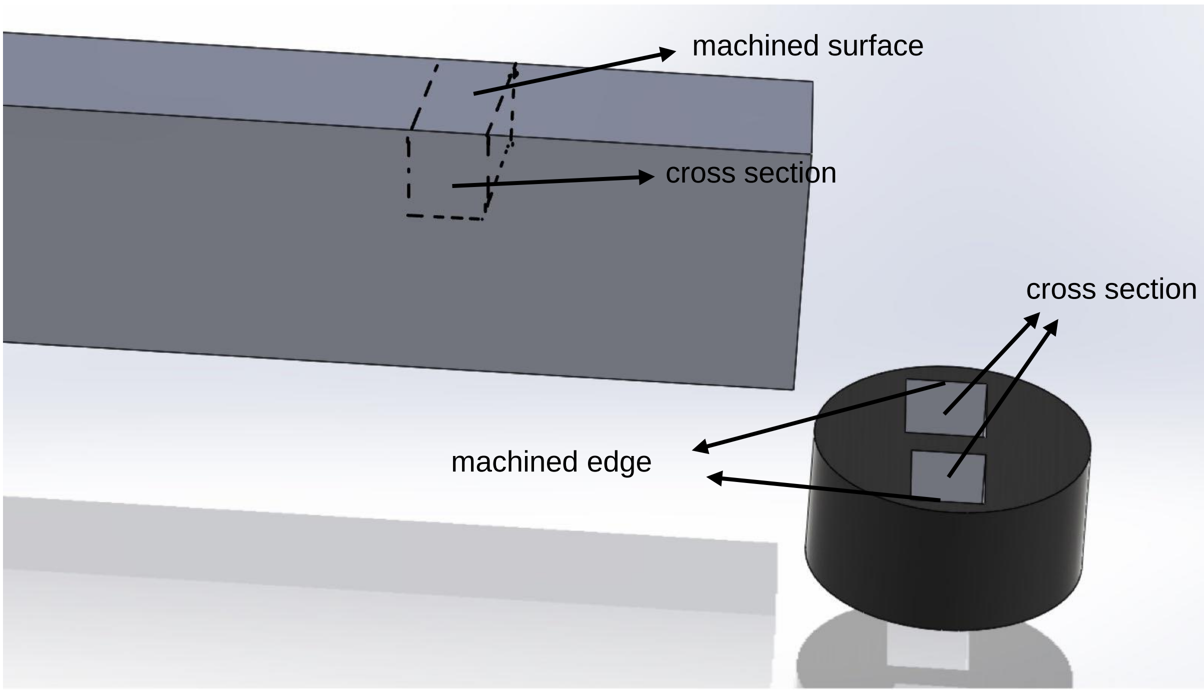


Figure 19. Scheme of sample preparation.

Since sample surfaces often have irregularities, scratches or surface defects, grinding and polishing are required to achieve a smoother surface for more accurate observations and results. After hot mounting, the mounted samples were put on the struers TegraForce-5 grinding and polishing equipment (as shown in figure 20) for grinding and polishing. In this process, different tray of a mill and suspensions were used to control the polishing process due to the varying depths of scratches and defects. After polishing with 9 µm, 6 µm and 1 µm suspensions, respectively, colloidial silica was used for finishing (as shown in figure 21). The finished prepared sample are shown in figure 22.

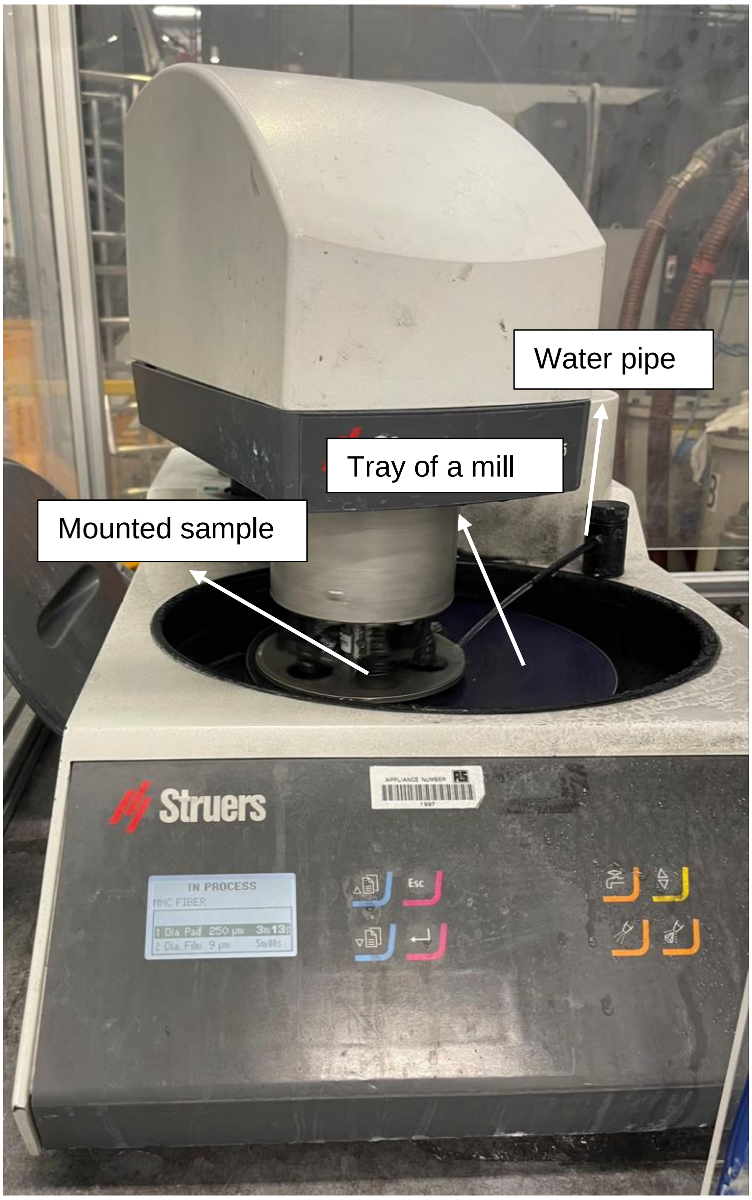


Figure 20. Struers TegraForce-5

(a) (b)

Figure 21. (a) Diamond suspensions (1µm, 9µm, 6µm from left to right); (b) Colloidial Silica.

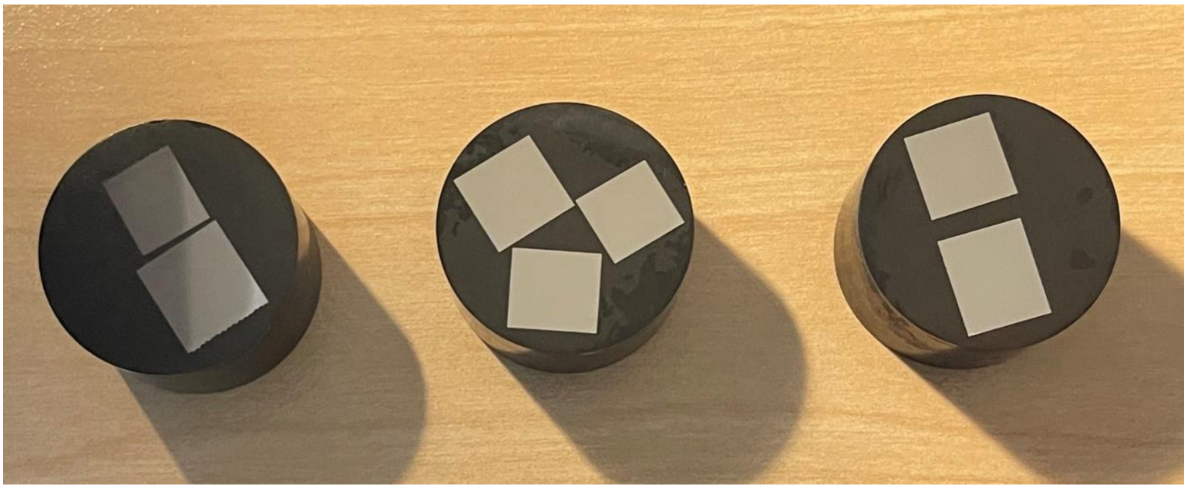


Figure 22. Prepared samples.

After the preparation of the samples, an initial examination of the samples was carried out using SEM (XL30) , but no significant microstructure could be observed, so the prepared samples were etched (a process used in material science and metallurgy to reveal the microstructure of a material by selectively removing layers of the material's surface through chemical or electrochemical means) for about 10 second using kroll's reagent (100 ml water, 1-3 ml hydrofluoric acid and 2-6 ml nitric acid).

The prepared Ti64 samples were re-examined after etching using SEM (XL30) and more pronounced microstructures were observed (as shown in figure 23). The samples were subsequently analysed using SEM (XL30) and Energy dispersive X-ray Spectroscopy

[redacted]

(EDS: oxford instrument X-Max) to investigate the effect of different processing methods, in particular LAMill, on the subsurface of the titanium alloy including the microstructure of the machined surface and the chemical component. The detailed results are presented in the results and discussion section.

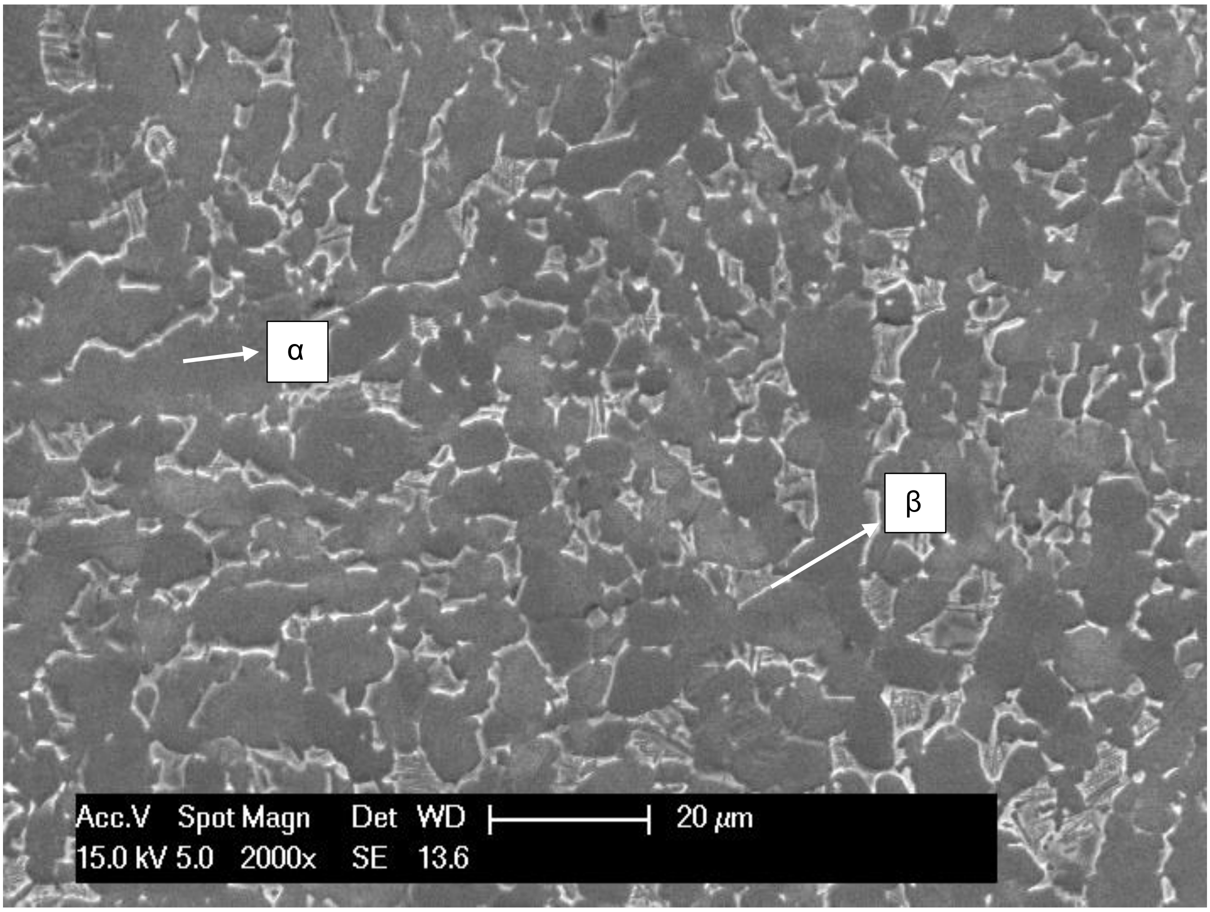


Figure 23. Microstructure of the prepared ti64 samples.

[redacted]

4. RESULTS AND DISCUSSION

4.1 The effect on cutting process

4.1.1 Cutting force

In milling cutting, the cutting conditions have a great influence on the workpiece's machining surface quality and machining efficiency. Among them, cutting force is one of the most important influencing factors [16]. In the machining process, higher cutting force will lead to worse surface roughness. In addition, higher cutting forces can also lead to higher residual stresses and thus affect the stability of the part. Therefore, it is worthwhile to study the cutting forces under different machining methods, which can predict the changing condition of the surface integrity of Ti64 alloys.

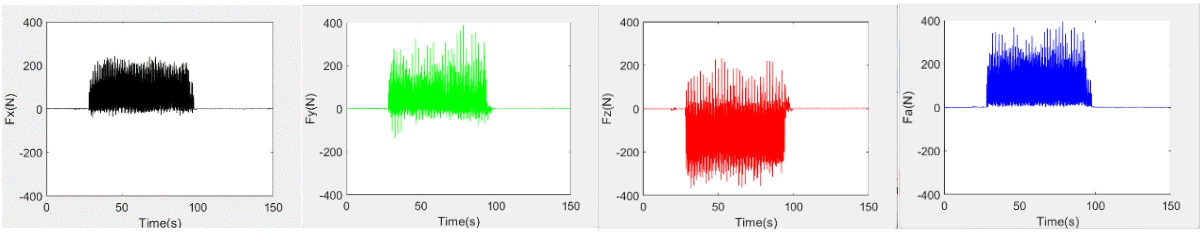
The results of average cutting force and maximum cutting force of CMill and LAMill of ti64 alloys were obtained from matlab and shown in table 6 and table 7 below. The waveform plots of the cutting force are also shown in Figure 24 for a more visual observation of the results.

Temperature (DOC)	Fx (N)	Fy (N)	Fz (N)	Fa (N)
RT	178.5917	147.2836	-233.8450	115.9776
650	20.8310	10.4066	-24.7696	29.1541
850	33.7380	21.7449	-44.7152	48.0289
1050	16.3614	7.9923	-21.6443	24.3785

Table 6. Average cutting force (N) in different directions.

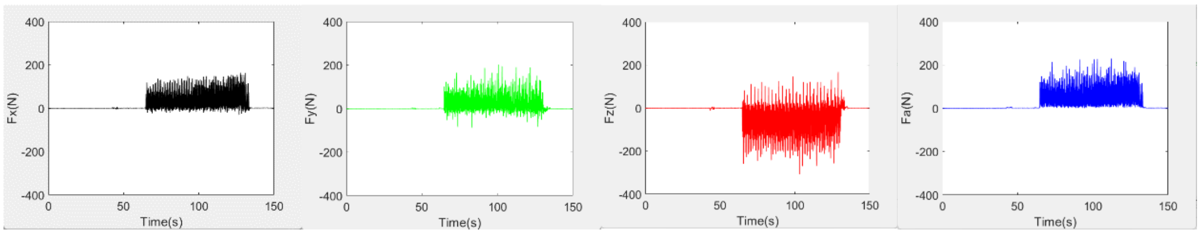
Temperature (DOC)	Fx (N)	Fy (N)	Fz (N)	Fa (N)
RT	257.4947	401.0315	-390.8571	410.2492
650	244.4740	344.2072	-378.3689	358.9350
850	244.1334	401.3147	-438.3354	418.8802
1050	260.4130	350.7210	-355.3009	382.4431

Table 7. Maximum cutting force (N) in different directions.

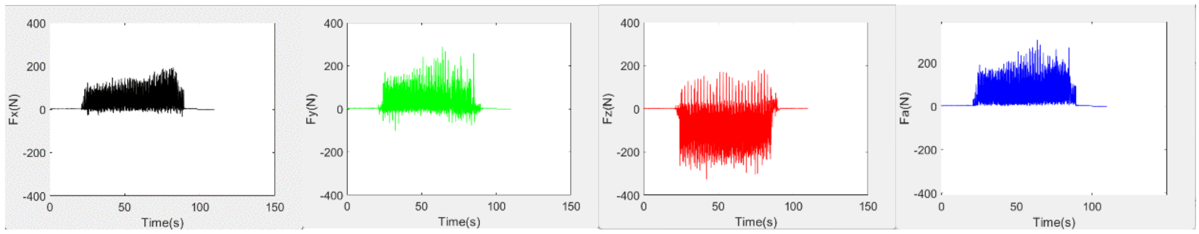


(a) (b) (c) (d)

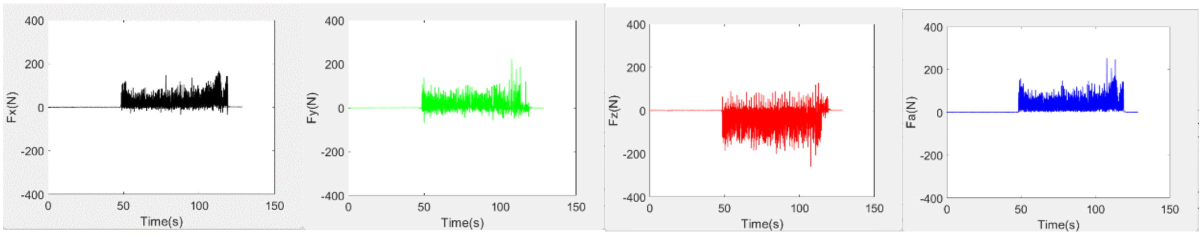
[redacted]



(e) (f) (g) (h)



(i) (j) (k) (l)



(m) (n) (o) (p)

Figure 24. Waveform plots of cutting force (N) under RT CMill: a), b), c), d); 650 °C LAMill: e), f), g), h); 850 °C LAMill: i), j), k), l); 1050 °C LAMill: m), n), o), p).

Since the feed direction is X direction and the cutting forces in the directions Y and Z are not important parameter for this research, the cutting force in X direction Fx and combined cutting force Fa were treated as primary parameters for analysis and comparison.

By comparing the shown cutting force results above, it was observed that the cutting forces in all directions with LAMill under three different temperatures were found to be reduced compared to CMill. However, there were differences in the results of cutting force changes under different temperature conditions. It was found that the average cutting force with 650 °C LAMill was reduced by about at least 100N compared to CMill in all the directions which may be due to the fact that the workpiece is softened by the thermal effect of the laser heating process. When the temperature came to 850 °C, it was found that the combined force and the force in feed direction Fx all increased a little bit (about 15N) compared to the force at 650 °C, which may be caused by the phase transition within the titanium due to the further increase in temperature [13]. The β-phase produced by the increase in temperature increases the strength of the material compared to the alpha-phase.

It was noticeable that the combined cutting force and Fx have decreased again compared to the cutting force at 650 °C when LAMill under 1050 °C. This is due to the extreme thermal

[redacted]

effect of high temperatures, which causes the surface of the workpiece to soften too much and removes a lot of unnecessary material.

In addition to these, it can be observed from the plots that the magnitude of the change in cutting force increases as the laser power increases, this is because an increase in temperature changes not only the strength but also the modulus of titanium. The former is related to a reduction of cutting force and the latter affects the variation in cutting force amplitude.

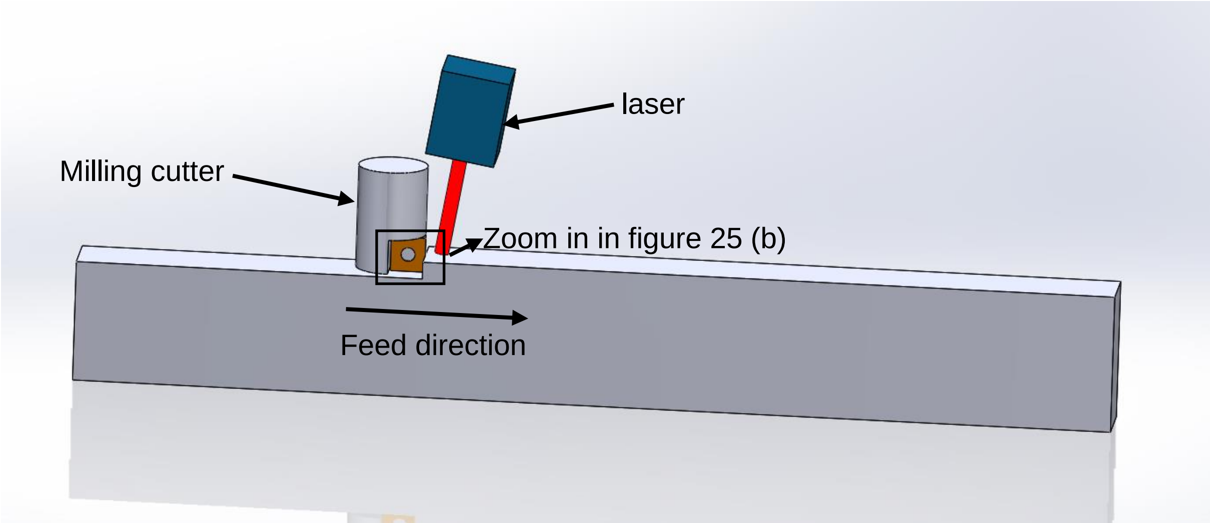
At the same time, it was found that during the CMill and the LAMill at 650 °C there was a slight vibration of the tool at the time of contact between the tool and the workpieces, whereas at 850 °C and 1050 °C this phenomenon was hardly observable, probably due to the fact that the thermal effect caused by the continuous increase in temperature sufficiently softened the material to be machined.

From these observations, it can be concluded that LAMill of Ti64 alloys can efficiently reduce the cutting forces in machining and thus achieve better machining results and herald better surface integrity. However, the temperature distribution on the surface of the workpiece during the machining process needs to be strictly controlled, otherwise the inappropriate laser power will cause excessive material removal and material waste.

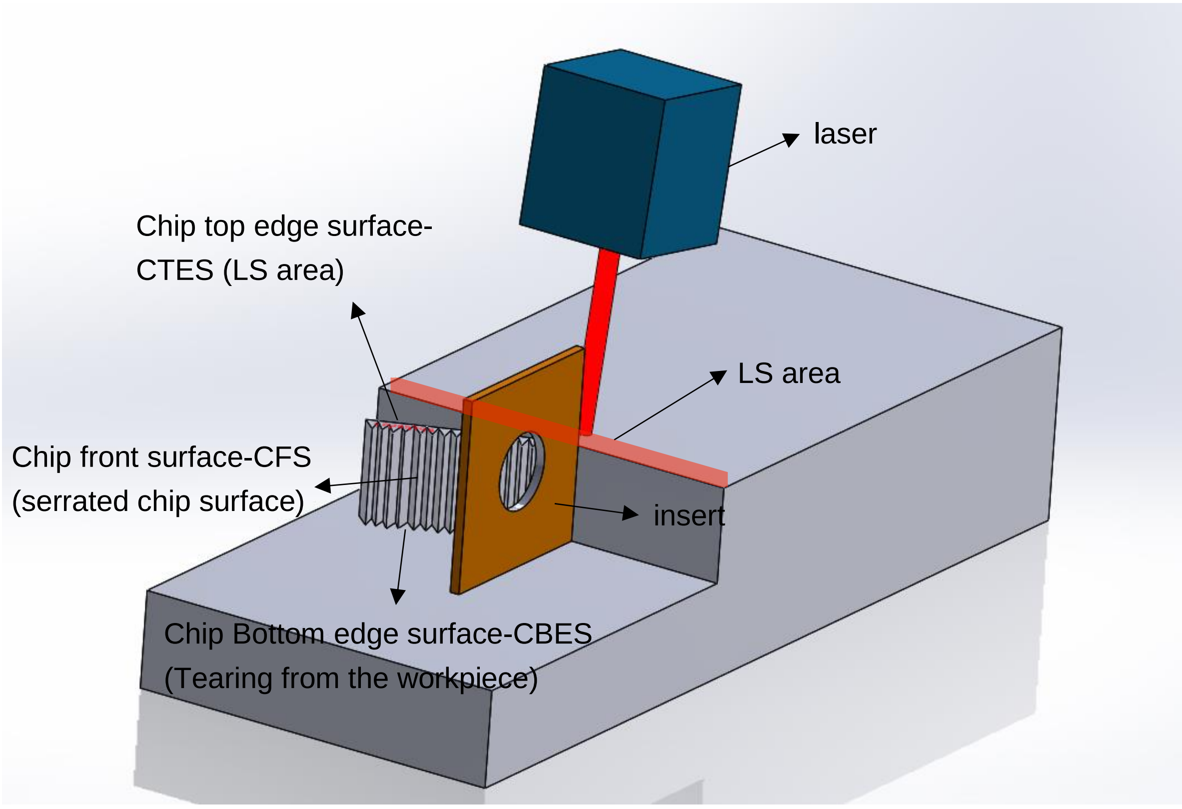
4.1.2 Cutting chips

The characteristics of the chips produced during the milling process can provide valuable information about the surface integrity of the material being machined. For instance, the shape of the chips can indicate whether the cutting conditions are stable; the colour of the chips can reflect whether the temperature during machining is appropriate; and the texture of the chips can reflect the interaction between the tools and the workpieces, and this information can be used to deduce whether the workpiece being machined has a good surface integrity or not. In LAMill, as the introduction of the laser fully heats and softens the material, the chips produced in this process should be expected to be completely different from those in CMill. Therefore, investigating the properties of chips, which are of great significance in the machining process, can contribute to the understand of the effects of the introduction of the laser and thus explore the impact on the surface integrity of the machined Ti64 alloys.

In order to fully understand the mechanism of the generation of the chips during milling of Ti64 alloys and the characteristics of the generated chips, as shown in figure 25, the generation of chips during milling and the definition of different regions of the chips were modelled and simulated using solidworks. The figure 25 (b) illustrated the definitions of different regions of the chips: the chip top edge surface (CTES) is the edge surface treated with the LS, the chip front surface (CFS) is the serrated chip surface and the chip bottom edge surface (CBES) is the surface which teared from the workpiece.



(a)



(b)

Figure 25. illustrate of the chip generation during milling process, a) overview of LAMill process; b) definition of the different chip area.

To investigate the differences between the chips with CMill and LAMill, the overall morphology of the chips was compared first. As shown in figure 26, it can be observed that the chips generated under the LAMill are generally less twisted than the chips generated under the CMill, and the chips became flatter with the increasing temperature. This phenomenon shows that the softening of the material by the laser in LAMill was effective, it reduced the torsional forces received by the material, which corroborates the reduction in cutting forces mentioned earlier. The reduction in chip torsion also indicates that the laser energy reduced the friction between the tool and the workpiece, resulting in smoother chips.

In addition, it was found that the burrs on the CBES of the chips produced under LAMill are dense and regular while the burrs on the CBES of the chips generated under CMill are larger, less and more irregular, as shown in figure 26 a) and c). This demonstrates the higher cutting forces and more severe material deformation inherent in CMill. In contrast, the denser and more regular burrs in LAMill reflect the fact that the combined effect of laser energy and mechanical cutting promotes finer material removal and chip formation characteristics, which also suggests a more homogeneous material response to the machining process in LAMill.

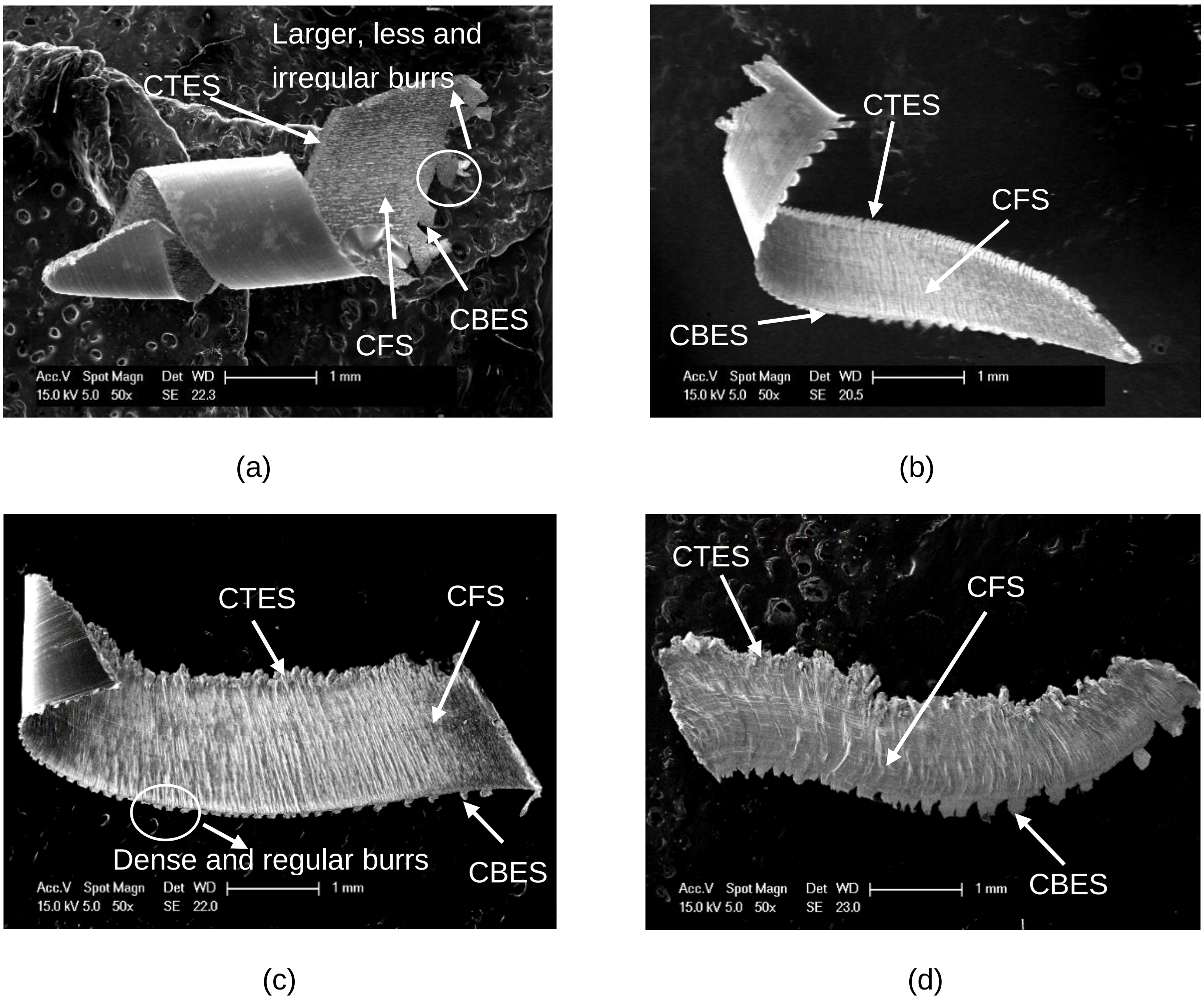


Figure 26. The overall morphology of the cutting chips under a) RT CMill; b) 650 °C LAMill; c) 850 °C LAMill and d) 1050 °C LAMill.

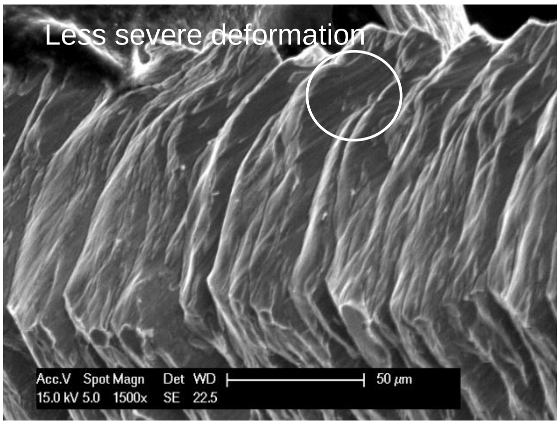
In order to further study the effect of the introduction of laser on the microstructure of the cutting chips. SEM observations with higher magnification on both CTES and CBES were implemented which are shown in figure 27 and figure 28 respectively. The reason that the

CFS of the chip was not analysed separately is that the texture of the CFS is essentially the same as that of the CBES.

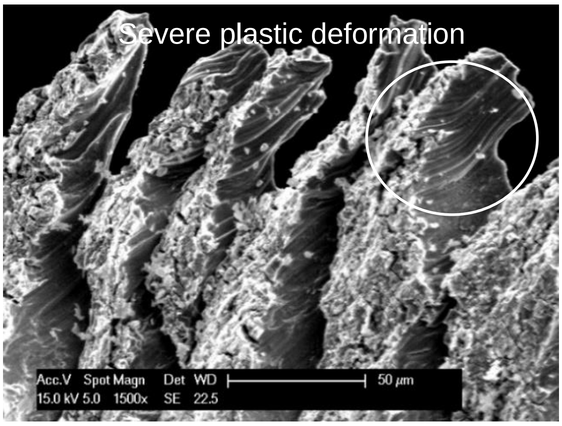
From the results of CTES shown in figure 27, it can be observed that due to the heating of the ti64 workpiece by the introduction of the laser, the honeycombed melt-like layer was formed on the CTES under the LAMill process (as shown in figure 27 d)). And the higher the temperature, the more pronounced the damage to the workpiece. The CTES of the chip at room temperature, on the other hand, shows a more regular tooth-like arrangement, with no obvious thermal effects, as shown in figure 27 a). Surprisingly, more severe grain slippage was observed on the CTES of the LAMill chips (as shown in figure 27 a) and c)), which is due to the fact that despite the reduction in cutting forces under LAMill, the CTES receives a strong influence from the thermal effect of the laser energy, and the sudden increase in temperature reduces the strength of the grain boundaries thus making them easier to displacements and deformations.

When it comes to the results of CBES, it was observed from the figure 28 that the CBES of the chips were significantly smoother under LAMill than the chips produced in CMill. As the temperature increases, the surface morphology of CBES gradually changes from an undulating tooth-like distribution to a flatter, continuous distribution. The change of the texture shows that due to the introduction of the laser, the material is fully softened, making the contact between the tool and the Ti64 alloy workpiece to be machined more stable, and providing less cutting force.

From these results above, it can be seen that the chips are flatter and less distorted with LAMill. Although the introduction of the laser produced more severe damage to the CTES of the resulting chip, since the CBES of the chip is the surface that is connected to the surface of the machined workpiece and the CFS and CBES of the chips under LAMill were smoother than the CMill. It can indicate that the LAMill should provide a better machined surface integrity compared to CMill. To further verify the results, the inspection and analysis of the machined surface were done and are presented in detail in section 4.3.



(a)



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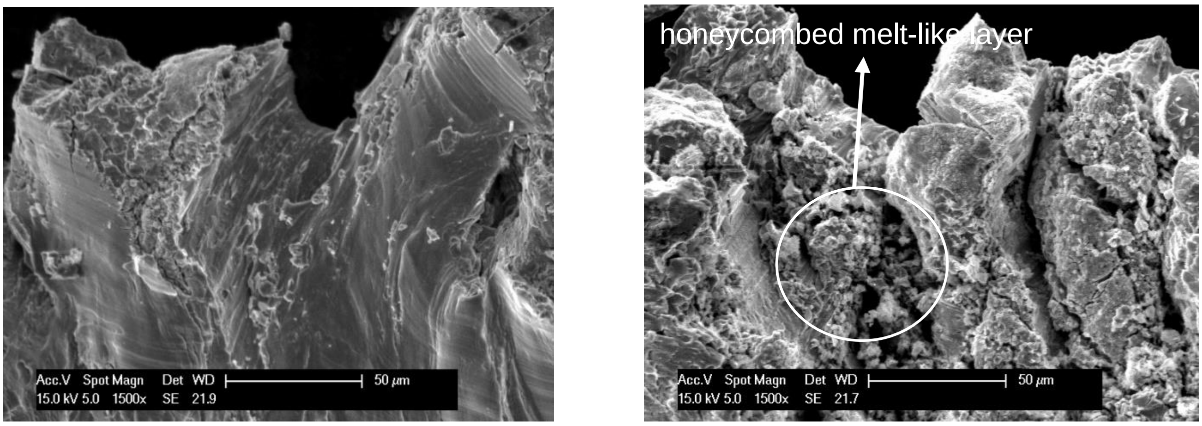
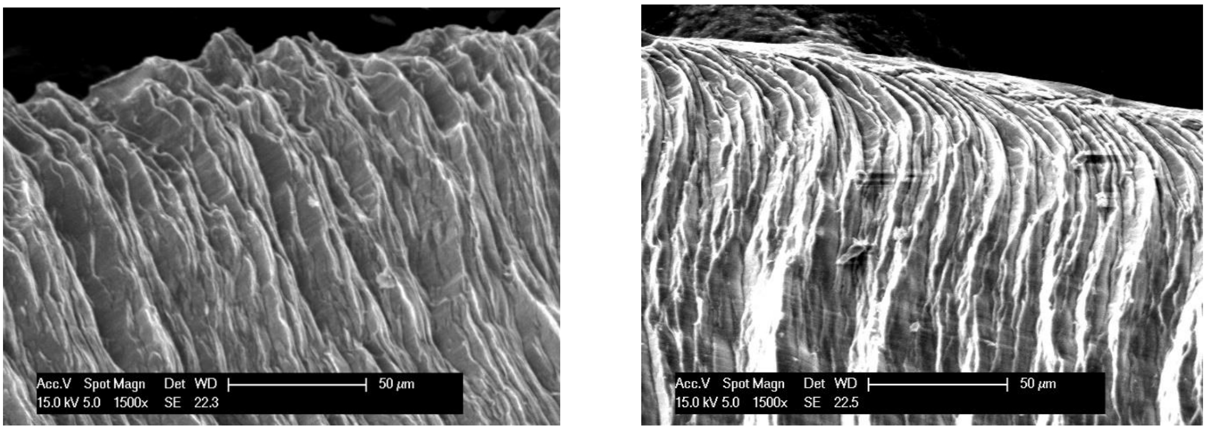
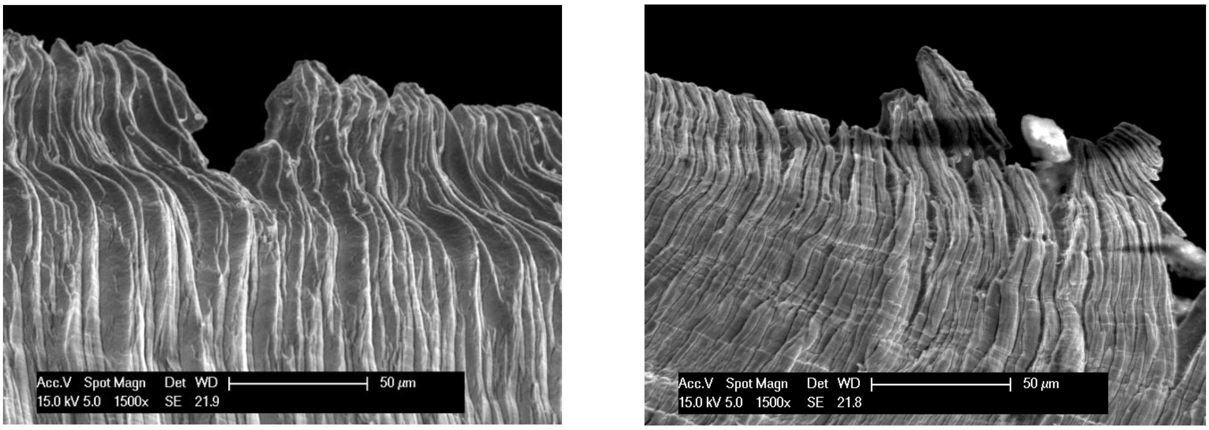


(c) (d)

Figure 27. CTES of the cutting chips under a) RT CMill; b) 650°C LAMill; c) 850°C LAMill and d) 1050°C LAMill.



(a) (b)



(c) (d)

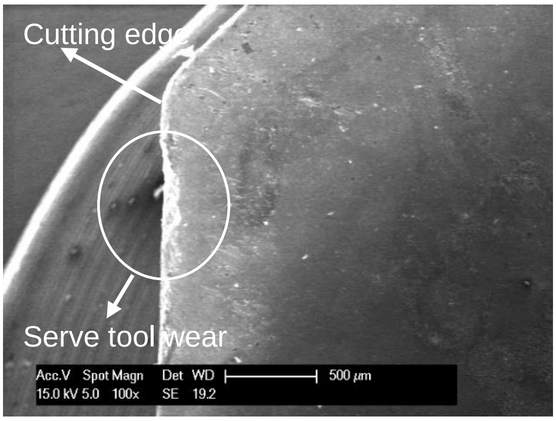
Figure 28. CBES of the cutting chips under a) RT CMill; b) 650°C LAMill; c) 850°C LAMill and d) 1050°C LAMill.

4.1.3 Tools wears

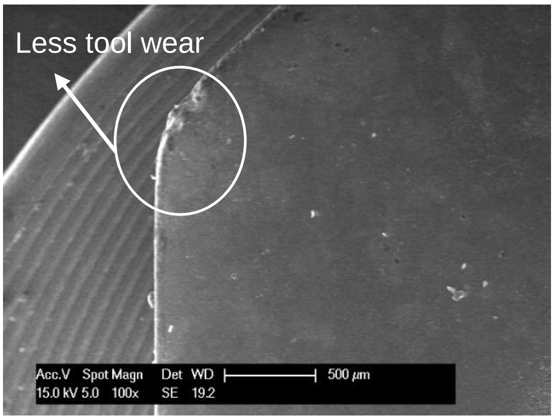
In milling process, tool wear can reflect cutting forces, cutting edge quality, and cutting stability during the cutting process. Therefore, the tool wear condition can laterally reflect the cutting quality in machining, and thus the impact on the surface integrity of the machined surface.

In order to comprehensively study tool wear in both CMill and LAMill and compare the differences, abrasion wear and chipping wear were investigated in these two machining methods. Abrasive wear usually occurs when a material comes into repeated contact with another surface, and chipping wear occurs when small fragments or chips of the material are dislodged from the surface due to localised stress or impact [9].

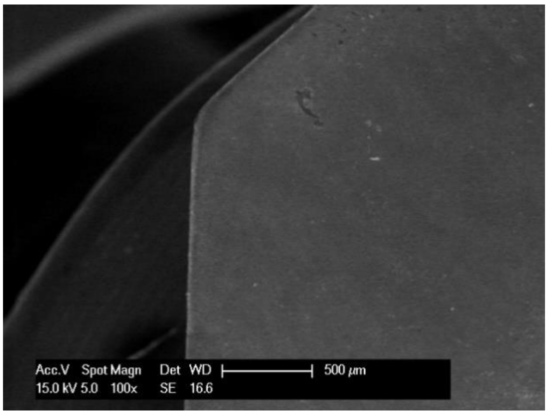
Firstly, the overall wear of the cutting edges of the tool was observed. From the results shown in figure 29, it can be observed that while the CMill process has caused more severe tool wear, the LAMill results in significantly less tool wear, especially at 850 °C and 1050 °C. This phenomenon can be explained by the fact that the thermal effect of laser heating in LAMill softens the material very well, thus reducing tool wear, which also confirms to a certain extent the reduction of cutting forces and a more stable cutting process.



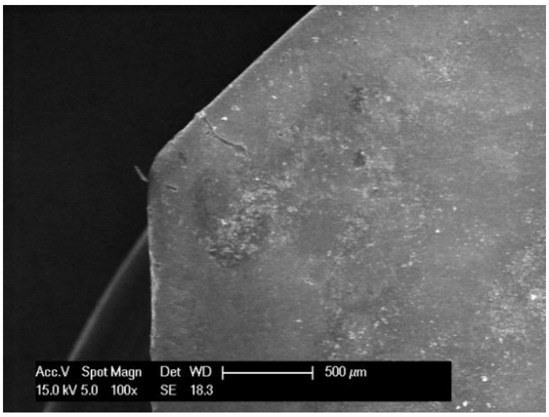
(a)



(b)



(c)



(d)

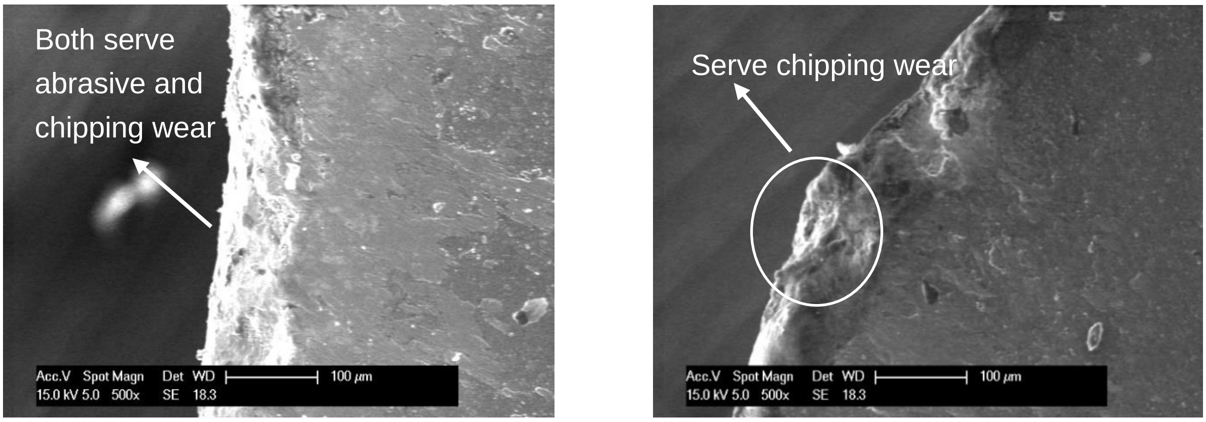
[redacted]

Figure 29. Overview of the tool wear under a) RT CMill; b) 650 °C LAMill; c) 850 °C LAMill; and d)1050 °C LAMill.

In order to study the variation of abrasive wear and chipping wear, higher magnitude results were taken for analysis. From the more detailed figures presented in Figure 30, it can be seen that at room temperature, the CMill caused both severe abrasive wear and chipping wear. Once the temperature reached 650°C, the abrasive wear condition had improved dramatically, although there was still more severe chipping wear. Since abrasive wear usually occurs due to friction between materials, the softening of the material caused by the laser heating undoubtedly reduced the friction. Nevertheless, in the previous study in 4.1.2, it was found that chips at 650 °C were still more curly and had fewer melted like- layers compared to the chips at 850 °C and 1050 °C. The chips generated during the cutting process inevitably caused chipping wear on the workpiece.

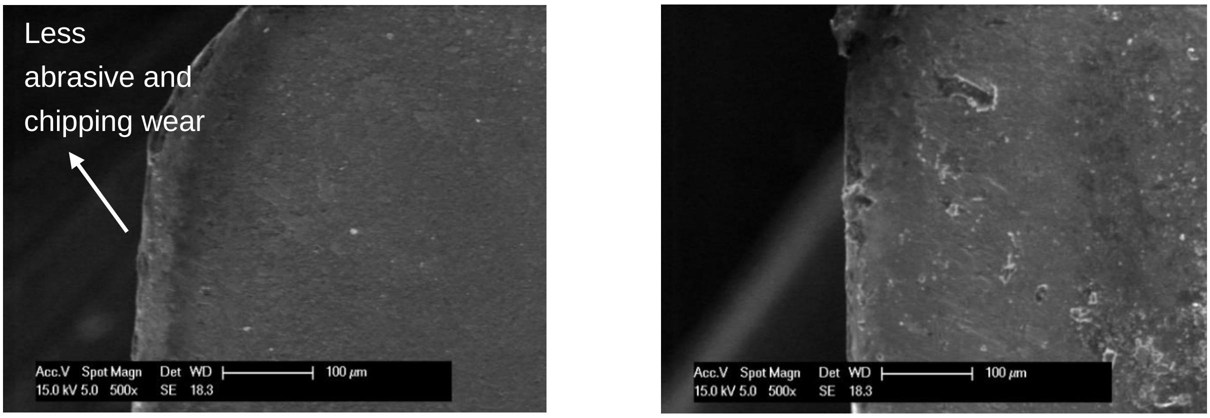
When the temperature reaches 850 and 1050 °C, both abrasive wear and chipping wear are significantly reduced compared to room temperature. Compared to 650 °C, the chips at 850°C and 1050°C are flatter, so the collision between the chips and the workpiece is weaker, reducing chipping wear.

In summary, LAMill can provide less tool wear to varying degrees compared to CMill of Ti64 alloys, which suggests the reduced cutting forces during the cutting process and a more stable cutting process with less tool vibration. Thus, the improvement of the surface integrity of the machined Ti64 alloy should be expected.



(a) (b)

[redacted]



(c) (d)

Figure 30. Details of the tool wear under a) RT CMill; b) 650 °C LAMill; c) 850 °C LAMill; and d)1050 °C LAMill.

4.2 The effect on machined Ti64 surface

4.2.1 Surface roughness

Surface roughness refers to the irregularities and deviations that exist on the surface of a material. Such surface defects are usually caused by machining methods, wear and tear, erosion, and so on. It can visualise the surface integrity of the machined part. Also, surface roughness can effectively affect the quality and assembly of parts [15]. Due to the fact that the ti64 alloy is usually used in the key parts of some precision instruments. It is valuable to study the effect of different machining methods on the surface roughness of Ti64 alloy.

To analysis the surface roughness, the average roughness (Ra) and the max value of surface peak to valley (Rt) were measured to analysis the surface roughness which represent the arithmetic average of the absolute values of the deviations from the mean line of the surface profile and the difference in height between the highest peak and the lowest valley within a measured surface profile. In addition, the softening of the Ti64 workpiece by the laser predicts a very different surface morphology, and therefore the surface morphology under different processing methods was also compared.

CMill VS Pure LS:

By comparing the Ra after the process of pure LS to CMill. It was found that the pure LS has increased the value of Ra and Rt significantly, and the higher the temperature, the more pronounced the change (as shown in figure 31 (a)), which can be contributed to the increasing temperatures causing increasingly severe thermal effects thereby producing higher material removal rates.

The further analysis of the surface morphology confirms this view. By comparing the surface morphology of the workpiece under pure LS and CMill. It was found that the surface morphology of the workpiece under LS was severely deformed. The surface morphology of

[redacted]

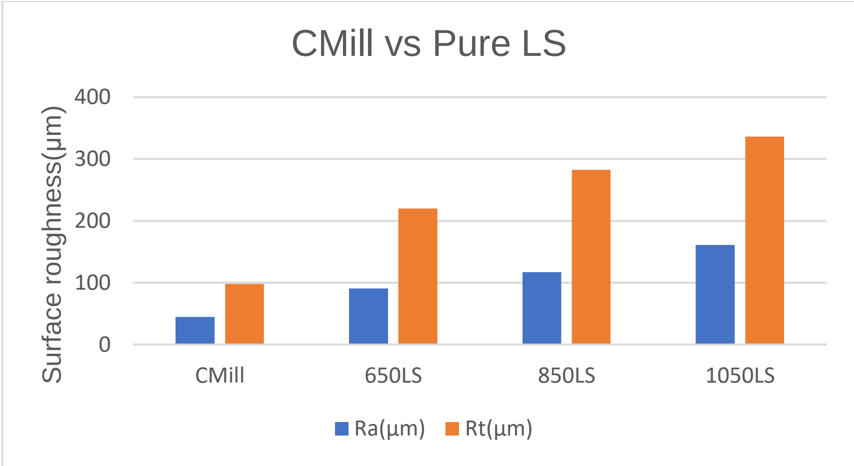
CMill is heavily influenced by the milling tool and exhibits a dense, fan-shaped cutting texture similar to the shape of the milling tool (as shown in figure 32 (a)). For the pure laser scanned workpiece, despite the fact that the workpieces were pre-treated with milling cuts before LS, no traces of tool cuts could be observed at all after LS. The surface morphology after LS shows a wave-like distribution of rounded and raised ridges as well as grooves and the depth and width of the grooves increased with increasing temperature (as shown in figure 33). The reason for this phenomenon is that the LS creates a heat affected zone (HAZ) on the surface of the workpiece, where the melting and recondensation of the material changes the surface morphology of the workpiece, and as the temperature increases, the more severe thermal effects lead to the removal of more material, resulting in a worse surface roughness.

CMill VS LAMill:

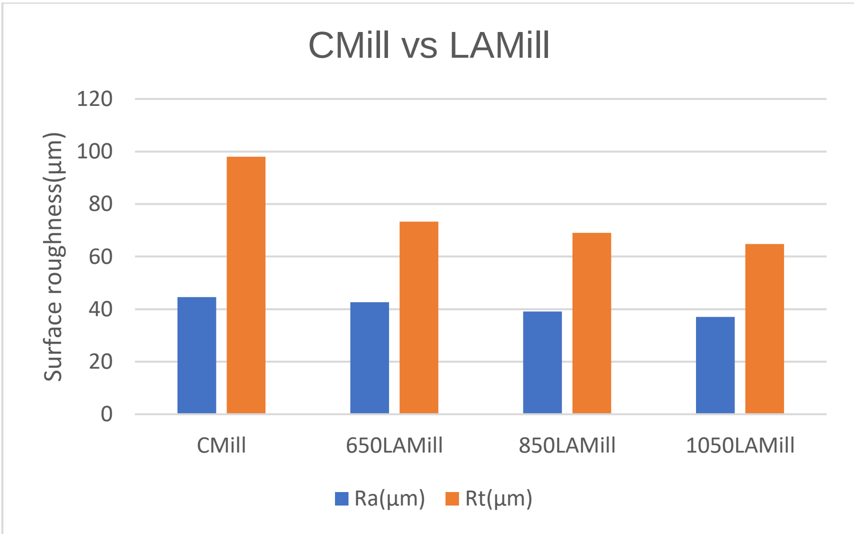
Surprisingly, the surface roughness results under LAMill show the exact opposite of pure LS both in value and surface morphology. The Ra of the workpiece as well as the Rt are reduced under LAMill, and this effect is more pronounced as the temperature increases (as shown in figure 31 (b)). These results are indicative of the fact that LAMill can indeed reduce the surface roughness in CMill of ti64 alloys and thus improve the surface quality as expected.

About the surface morphology, there were no significant changes in surface morphology were observed under LAM. Although the ti64 workpieces were also heated by the laser in LAMill, it did not show the same surface depressions and bulges as under pure LS. The surface morphology under LAMill is still similar to that under CMill, i.e., a dense, prop-like, fan-shaped texture dominated by tool influence (as shown in figure 32 b); c) and d)). However, the surface of the workpiece under LAMill has some obvious black traces left by the laser scanning and the number of grooves is reduced. In addition, the surface of the workpiece under CMill has many black voids, while the surface under LAMill still has these small holes, but the number and area of these holes are significantly reduced, which also predicts an improvement of the surface roughness.

From the above results, it can be shown that pure LS alone can greatly damage the workpiece but LAMill of titanium alloys can really reduce the value of surface roughness after machining compared to the CMill and thus improve the surface quality.

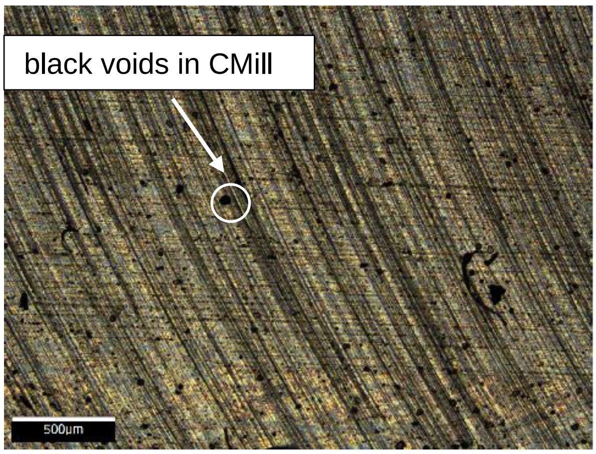


(a)

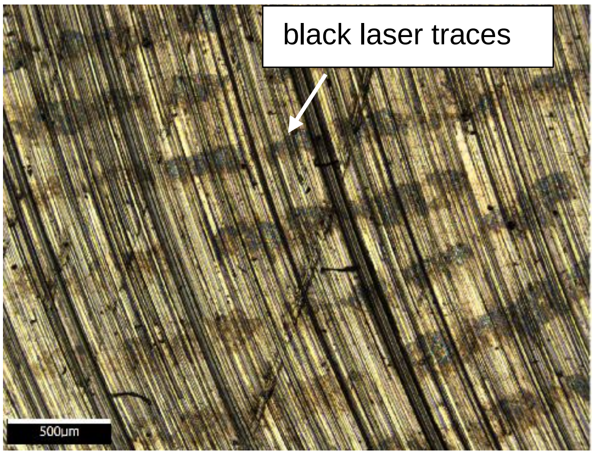


(b)

Figure 31. Comparison of Ra (µm) and Rt (µm), a) CMill vs LS; b) CMill vs LAMill.

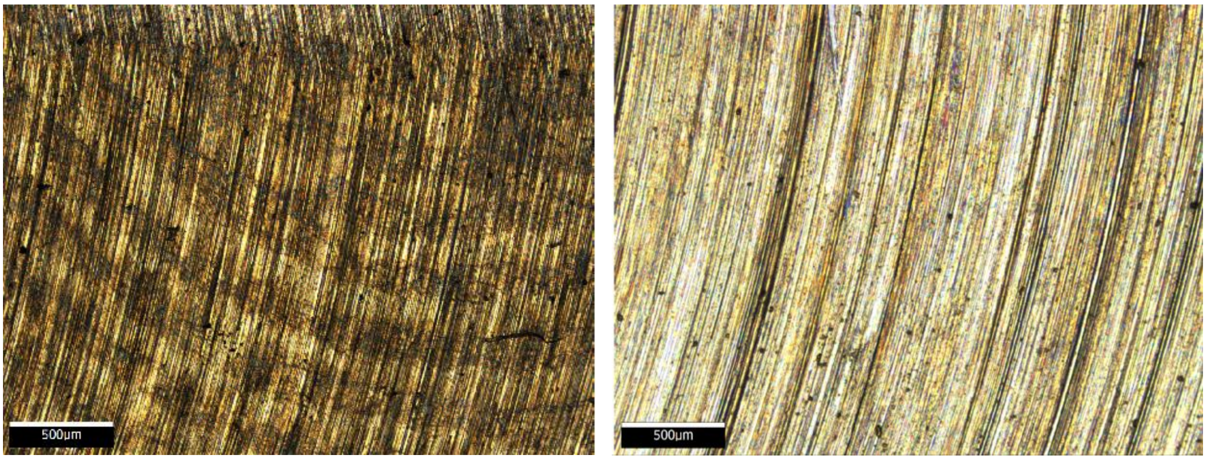


(a)



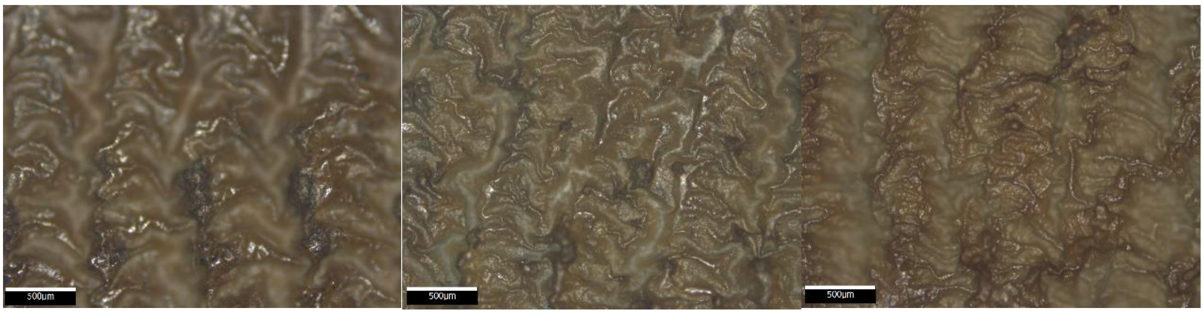
(b)

[redacted]



(c) (d)

Figure 32. Ti64 workpiece surface morphology under a) CMill; b) LAMill at 650 °C; c) LAMill at 850 °C; d) LAMill at 1050 °C.



(a) (b) (c)

Figure 33. Ti64 workpiece surface morphology under LS, a) 650 °C; b) 850 °C; c) 1050 °C.

4.3 The effect on machined Ti64 subsurface

4.3.1 Microstructure analysis of the machined workpiece surface integrity

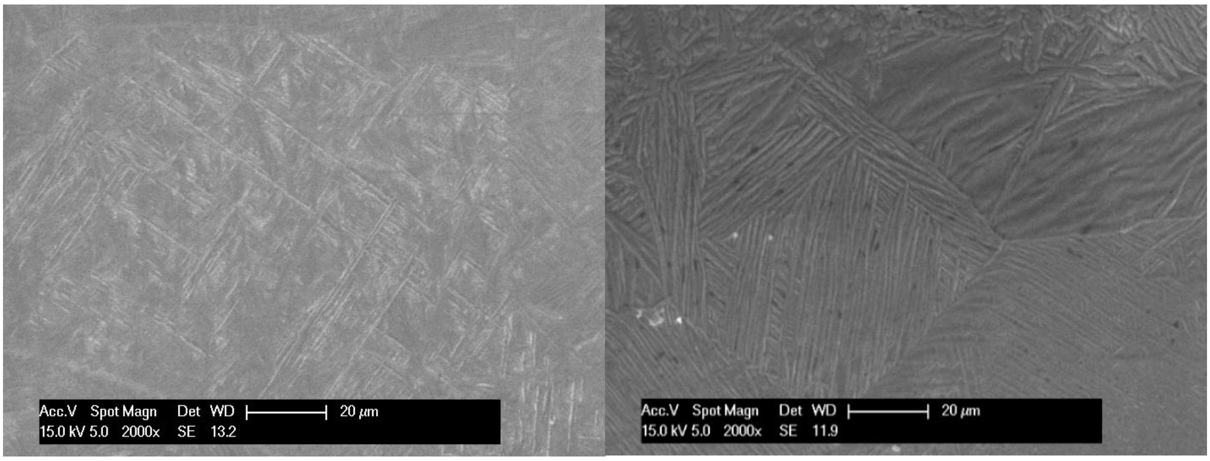
In order to further explore the different variations in the surface integrity of titanium alloys, the sub surfaces of the machined layers were analysed and compared separately for the different machining methods.

LS vs LAMill:

Firstly, the sub surfaces under LAMill and LS were compared to investigate whether the introduction of laser during milling would have a damaging effect. After comparing the phase transitions on the near-machined surfaces, it was found that the phase transitions under LAMill were not as pronounced as those under pure LS. For instance, as shown in Figure 34 b) and a) respectively, the phase of the near-machined surface under pure LS at the same 1050 °C shows a clear lamellar alpha + beta with alpha grain boundary while the phase under LAMill is alpha’ martensite and beta phase. This suggests that the novel control model

[redacted]

used during LAMill controls the temperature of the workpiece surface well, resulting in a more uniform surface temperature distribution. In contrast, pure LS could lead to locally higher temperatures, which are more likely to initiate phase transitions.



(a) (b)

Figure 34. Phase transition under the machined surface under a) 1050 °C LAMill; b)1050 °C LS.

By comparing the sub surfaces of the workpieces under LAMill and pure LS, it was found that a thick white layer and more severe edge effects (A phenomenon observed in electron microscope scans occurs when protrusions and edges of the sample's machined surface become very bright). were present under pure LS (figure 36 b), whereas less than 1 µm of edge effect layer was present under LAMill (figure 35 d). This difference may be due to the fact that pure LS is a non-contact process where the laser is applied directly to the metal surface, and therefore the metal surface may be more susceptible to high temperatures and high energies, resulting in a thick white layer and more severe edge effect. LAMill, on the other hand, is a contact machining method in which the laser is combined with a tool for machining, and therefore the metal surface may be subjected to the cutting action of the tool and coolant, which reduces the formation of the edge effect. In addition, severe damage to edges and grain enhancement during LS (figure 36 f)) did not occur in LAMill. Therefore, the introduction of laser in milling process does not have a damaging effect on the surface integrity of Ti64 alloys.

CMill vs LAMill:

Surprisingly, no obvious edge effect appeared on the subsurface of the machined face after CMill. Nevertheless, a comparison with the sub-surfaces after LAMill showed that the laser-assisted machined surfaces were significantly flatter and less severely damaged (figure 35 b) and d)). In addition, no clear phase transformation occurred under the machined surface of CMill, which indicates that thermal effects are not significant during CMill.

In summary, the introduction of laser into the milling process of Ti64 alloys does not have the same destructive effect on the machined surface as pure LS and improves the unevenness

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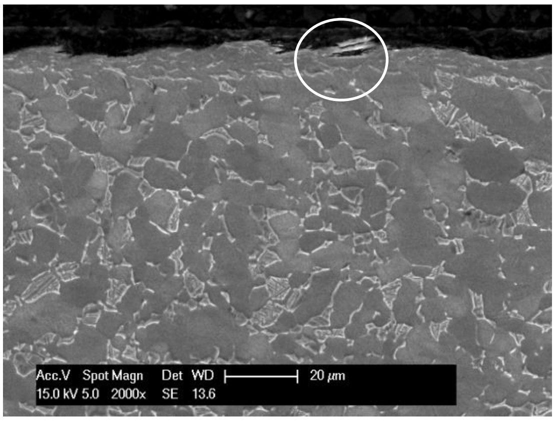
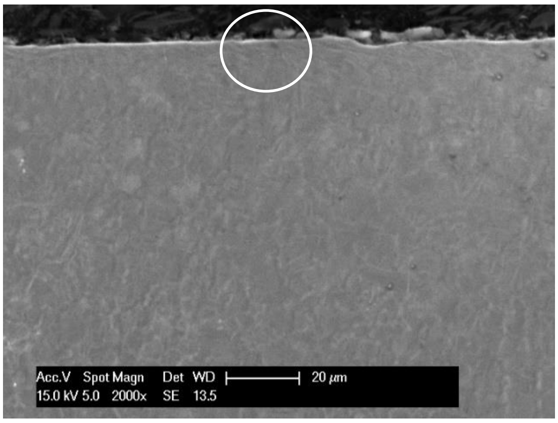
of the machined surface after CMill, which indicate that the LAMill of titanium alloys can really improve the surface integrity of titanium alloys after machining.



(a)

(b)



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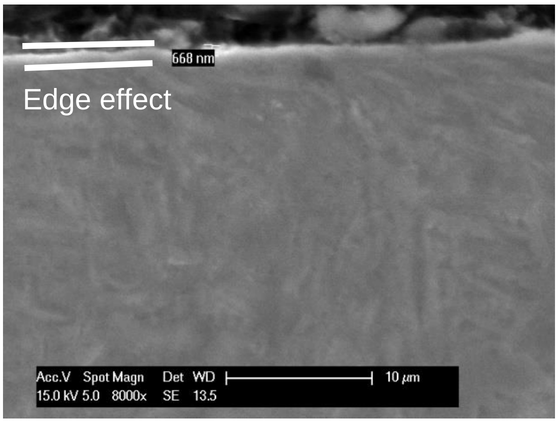
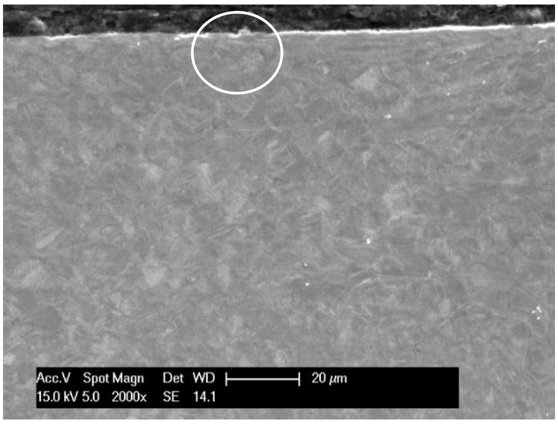
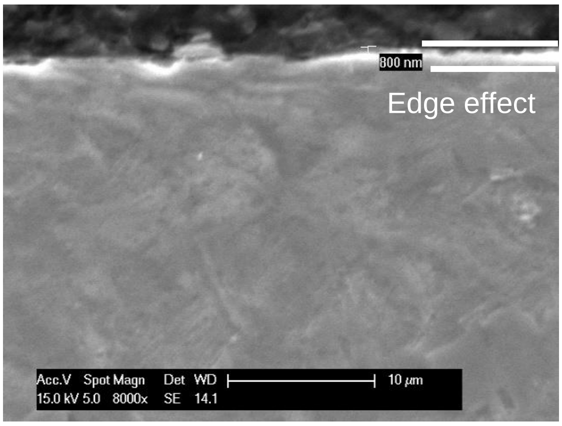


(d)

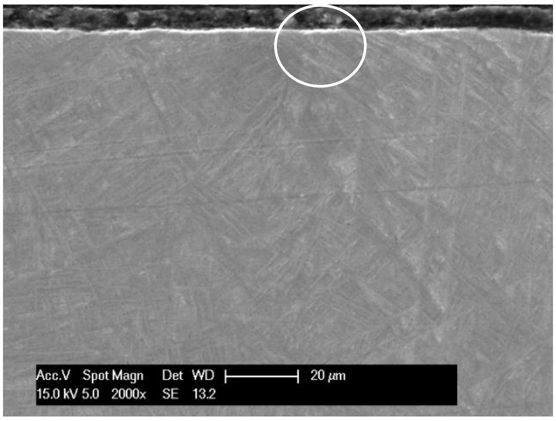


(e)

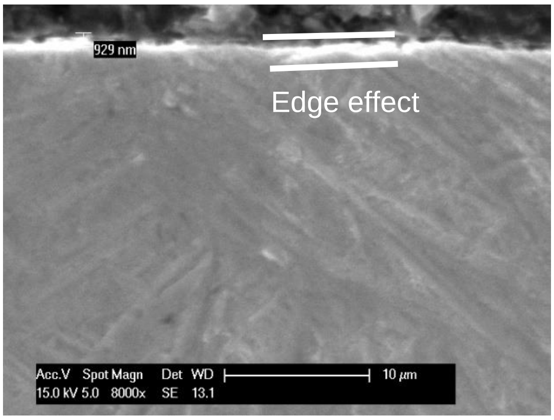


(f)

[redacted]

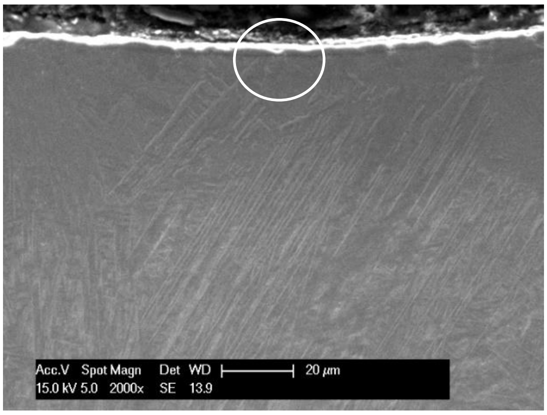


(g)

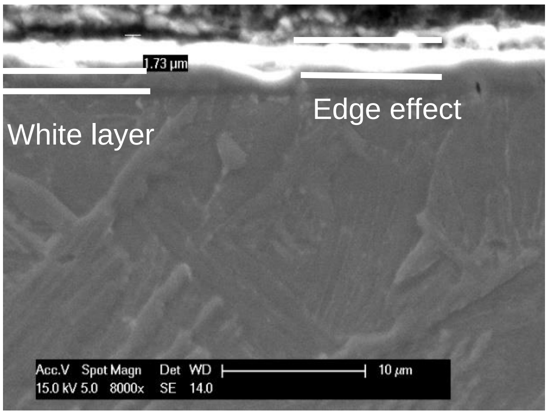


(h)

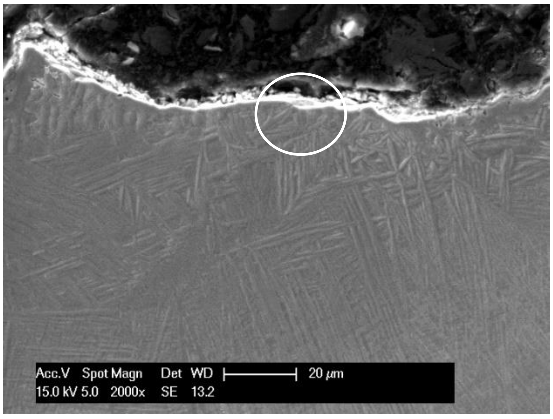
Figure 35. The mechanisim of the Ti64 workpiece machined surface under a),b) RT CMill ; c),d) 650 °C LAMill ; e),f) 850 °C LAMill ; g),h) 1050 °C LAMill.



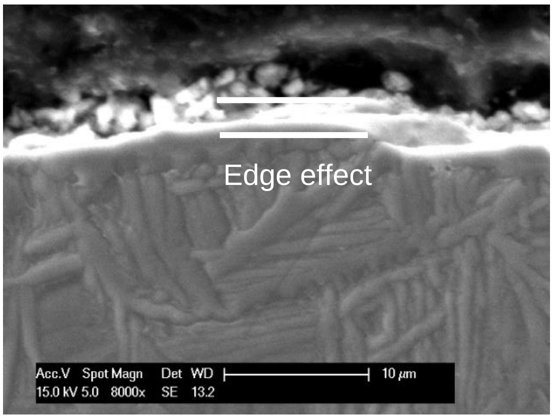
(a)



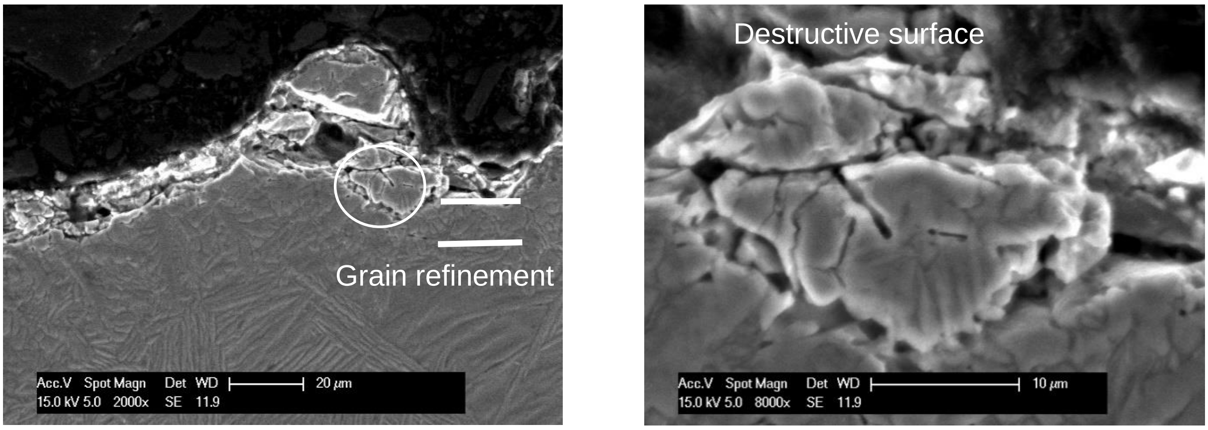
(b)



(c)



(d)



(e) (f)

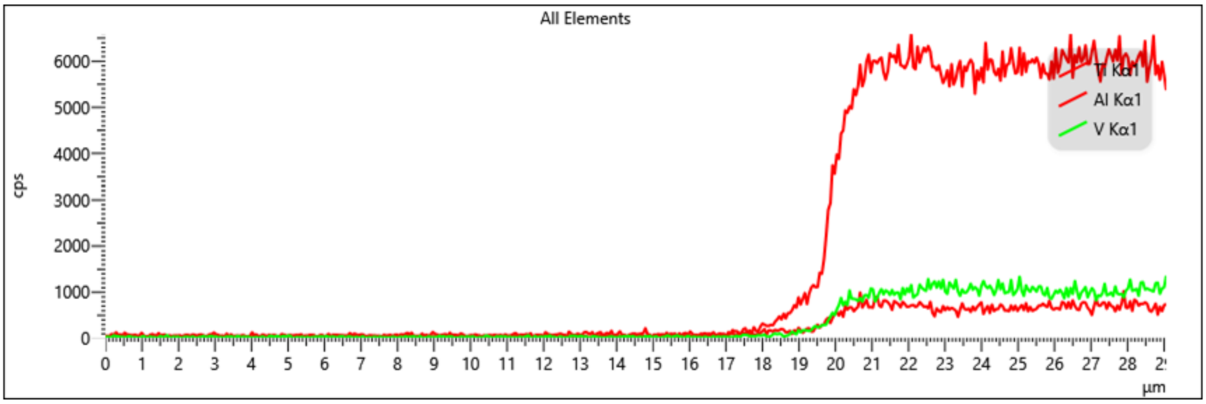
Figure 36. The mechanisim of the Ti64 workpiece machined surface under a),b) 650 °C LS ; c),d) 850 °C LS ; e),f) 1050 °C LS.

4.3.2 Chemical contents analysis of the machined surface

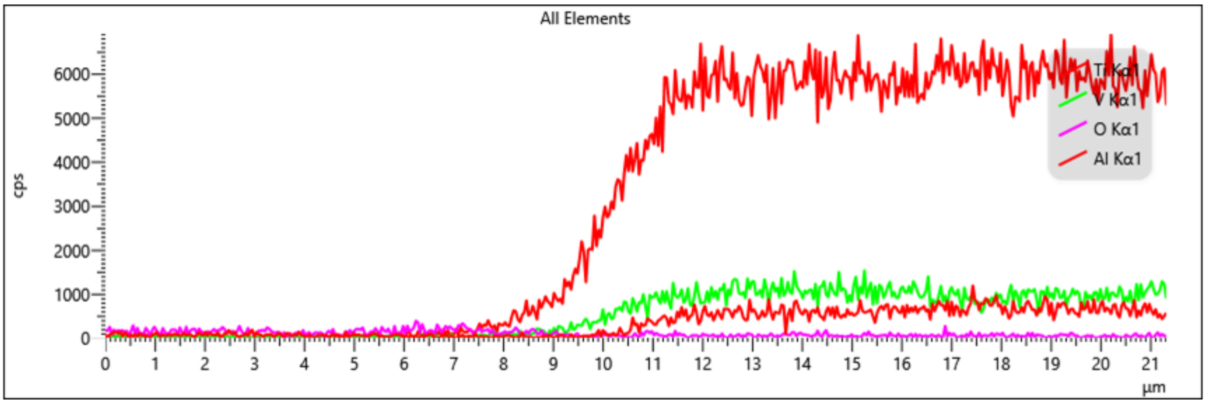
From the results in 4.3.1, it was found that the edge effect appeared on the machined surfaces after the introduction of the laser, both after pure LS and after LAMill, whereas it did not appear under CMill, and therefore further analyses of the chemical composition of the machined surfaces were required to investigate the specific reasons for the formation of the edge effect.

After analysing the chemical contents of the machined surfaces under different machining methods using EDS, it was found that elemental oxygen was produced on the surfaces after both LS and LAMill, which was not detected in CMill (as shown in figure 37). This is because the high energy density of the laser beam during laser processing leads to a rapid increase in the surface temperature of the material. At high temperatures, the material undergoes an oxidation reaction with the oxygen in the surrounding air, forming an oxide layer.

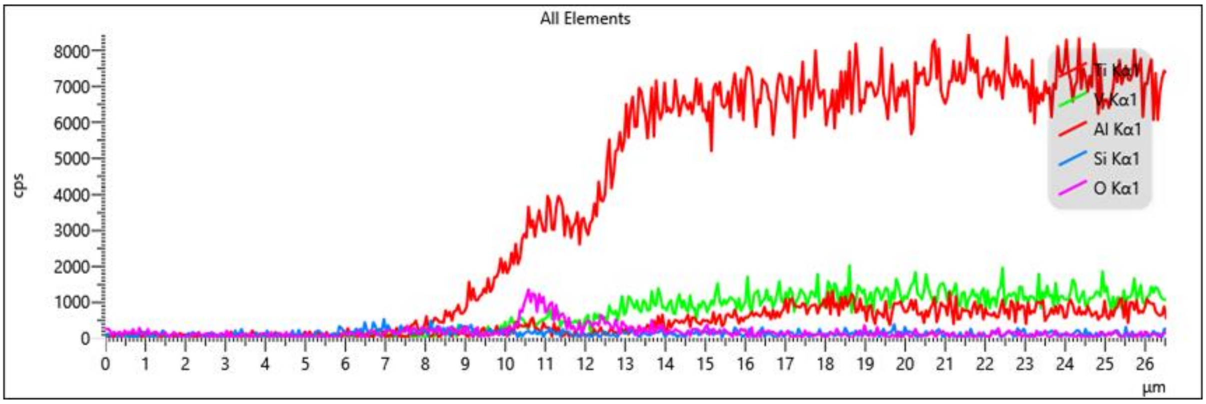
Thus, it can be deduced that the edge effect that appears after the introduction of the laser may be due to the presence of oxidation and chemical reactions in the heat affected zone caused by high temperatures.



(a)



(b)



(c)

Figure 37. EDX inspection of the machined surface under a) CMill; b) LAMill; c) LS.

4.4 Recommend parameters in LAMill of Ti64 alloy

From the results and analysis in section 4.1,4.2,4.3. It can be found that compared to CMill, LAMill of Ti64 alloys does improve post-machining surface integrity by reducing cutting forces, lowering tool wear, reducing surface roughness, and attenuating damage to machined sub surfaces. And the introduction of laser light during milling does not have the same destructive consequences as pure LS. However, it was also found that LAMill of titanium alloys at different temperatures resulted in differences in the degree of improvement in surface integrity. Therefore, it is necessary to select an optimal parameter based on the three temperatures applied in this experiment.

The experimental results described above show that in LAMill of ti64 alloys, the degree of improvement tends to increase with increasing temperature. However, it was also noticed that at the temperature of 1050°C, the material may appear to be excessively softened. For instance, the excessive reduction in cutting forces, honeycomb melted layers on the CTES of the chip and thicker white layers under the machined subsurface compared to the results under 650 °C and 850 °C. At 650 °C and 850 °C, the surfaces were not affected by the same excessive thermal effects in 1050 °C, however, because surface integrity was not improved

[redacted]

at 650 °C to the same extent as at 850 °C. Therefore, based on the experimental data, 850 °C should be the optimal parameter for LAMill of titanium alloys among 650 °C, 850 °C and 1050 °C.

CONCLUSIONS

In this experiment-based project, Ti64 alloys were subjected to CMill, LAMill and pure LS and the machined workpieces were cut and sampled. Observations and analyses of the machined workpiece and the prepared samples were carried out. It was found that LAMill of Ti64 alloys can improve the poor surface integrity of Ti64 alloys machined by CMill by reducing cutting forces, tool wear, surface roughness and sub-surface damage. In addition, the LAMill process did not show the same destructive surface results as in pure LS. However, as three different temperatures were used in the LAMill experiments, inconsistent improvements in surface integrity were found at different temperatures. By analysing the comparisons, it was found that the results at 850 °C were better than at 650 °C and did not show the thermal effects of the transition at 1050 °C. Therefore, 850 °C should be considered as the optimum temperature parameter for LAMill of Ti64 alloys. By applying this technology, it should be expected that the machining rate and durability of certain parts can be improved.

5. FUTURE WORK

In order to make this study to be more complete and more convincing, the study of the microhardness should also be conducted, since changes in the microhardness of a material are usually related to changes in the internal grain structure of the material as well as changes in surface roughness, it is of interest to investigate the effect of the introduction of laser light during the milling process on microhardness. However, as the machine for hardness testing was broken at that time (as shown in figure 38). Thus, this work should be done in future if possible.

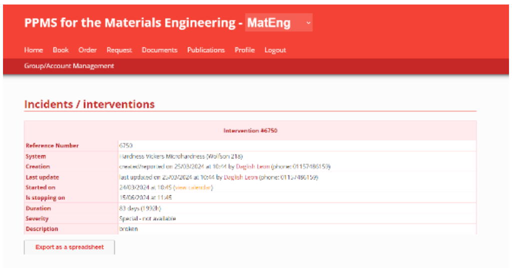


Figure 38. Incidents of the broken microhardness machine.

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7. ACKNOWLEDGEMENT

First of all, I would like to express my deepest gratitude to my supervisor, Dr Zhirong Liao, who always patiently answered my questions and offered valuable advice, and generously imparted his expertise! With his encouragement and help I have gained a deeper understanding in the field of materials science and advanced manufacturing, and his professionalism and diligence also inspire me at all times. His teachings will have a profound impact on my future academic and professional career.

I would also like to express my heartfelt thanks to PhD student [redacted] for teaching me lab techniques, patiently answering my questions and consistently offering constructive suggestions during the project!

In addition, I would like to thank Research fellow [redacted], Technician [redacted] and Technician [redacted]for their help and guidance in the experiments.

I would also like to thank all the professors who have taught and helped me, my personal tutors Dr Lee Harper and Dr Ming Li for their encouragement and care throughout my school life, and the University of Nottingham for providing me with a wealth of learning resources and an exciting undergraduate life.

APPENDIX 1 – Health and Safety documents

- PPMS for the MatEng Facility_L
- nmRC Access Agreement_Lianx
- Activity Risk Assessment Form
- [redacted]- nmRC Induction,

APPENDIX 2 – Other relevant material

```matlab
%Cutting FORCE Figure
clear
clc
close all

result = convertTDMS(true,'LAMTi_Carbide_Force_24_RT.tdms');
%%
ai0 = result.('Data').('MeasuredData')(3).('Data')/1e-2;
ai1 = result.('Data').('MeasuredData')(4).('Data')/1e-2;
ai2 = result.('Data').('MeasuredData')(5).('Data')/1e-2;

samf = 50000;
Caliai0 =0; Caliai1 =0; Caliai2 =0;

Fy = -(ai0+Caliai0);
Fz = (ai1+Caliai1);
Fx = ai2+Caliai2;


Fxlp = lowpass(Fx,0.015);
Fylp = lowpass(Fy,0.015);
Fzlp = lowpass(Fz,0.015);
Fa = sqrt(Fxlp.^2+Fylp.^2);

ForceData(:,1)=Fx;
ForceData(:,2)=Fy;
ForceData(:,3)=Fz;

ForceDatalp(:,1)=Fxlp;
ForceDatalp(:,2)=Fylp;
ForceDatalp(:,3)=Fzlp;
ForceDatalp(:,4)=Fa;
```

```matlab

T = table(Fx, Fy, Fz,'VariableNames', { 'Fx', 'Fy', 'Fz'} );
Tlp = table(Fxlp, Fylp, Fzlp, Fa, 'VariableNames', { 'Fx', 'Fy', 'Fz', 'Fa'} );

% Write data to text file
writetable(T, 'ForceData.txt');
writetable(Tlp, 'ForceDatalp.txt');


% ForceData(:,1)=Fx(100:1000000);
% ForceData(:,2)=Fy(100:1000000);
% ForceData(:,3)=Fz(100:1000000);

%xlswrite('ForceExcel.xlsx',ForceData);
save('ForceExcel','ForceData');
save('ForceExcellp','ForceDatalp');
t = (1:1:length(ai1))*1/samf;


downsample_factor = 4000;
t_downsampled = t(1:downsample_factor:end);
Fx_downsampled = Fx(1:downsample_factor:end);
Fy_downsampled = Fy(1:downsample_factor:end);
Fz_downsampled = Fz(1:downsample_factor:end);
Fxlp_downsampled = Fxlp(1:downsample_factor:end);
Fylp_downsampled = Fylp(1:downsample_factor:end);
Fzlp_downsampled = Fzlp(1:downsample_factor:end);
Fa_downsampled = Fa(1:downsample_factor:end);


figure(1)
plot(t_downsampled, Fx_downsampled, 'k', 'linewidth', 1)
```

```matlab
xlabel('Time(s)')
ylabel('Fx(N)')
set(gca, 'FontSize', 16)
xlim([0, 150])
ylim([-400 400])
saveas(gcf, 'Fx_downsampled.png')
Fx_downsampled_positive = Fx_downsampled(Fx_downsampled >50);
Fx_downsampled_average = mean(Fx_downsampled_positive);


figure(2)
plot(t_downsampled, Fy_downsampled, 'g', 'linewidth', 1)
xlabel('Time(s)')
ylabel('Fy(N)')
set(gca, 'FontSize', 16)
xlim([0, 150])
ylim([-400 400])
saveas(gcf, 'Fy_downsampled.png')
Fy_downsampled_positive = Fy_downsampled(Fy_downsampled>50);
Fy_downsampled_average = mean(Fy_downsampled_positive);

figure(3)
plot(t_downsampled, Fz_downsampled, 'r', 'linewidth', 1)
xlabel('Time(s)')
ylabel('Fz(N)')
set(gca, 'FontSize', 16)
xlim([0, 150])
ylim([-400 400])
saveas(gcf, 'Fz_downsampled.png')
Fz_downsampled_positive = Fz_downsampled(Fz_downsampled <-50);
Fz_downsampled_average = mean(Fz_downsampled_positive);
```

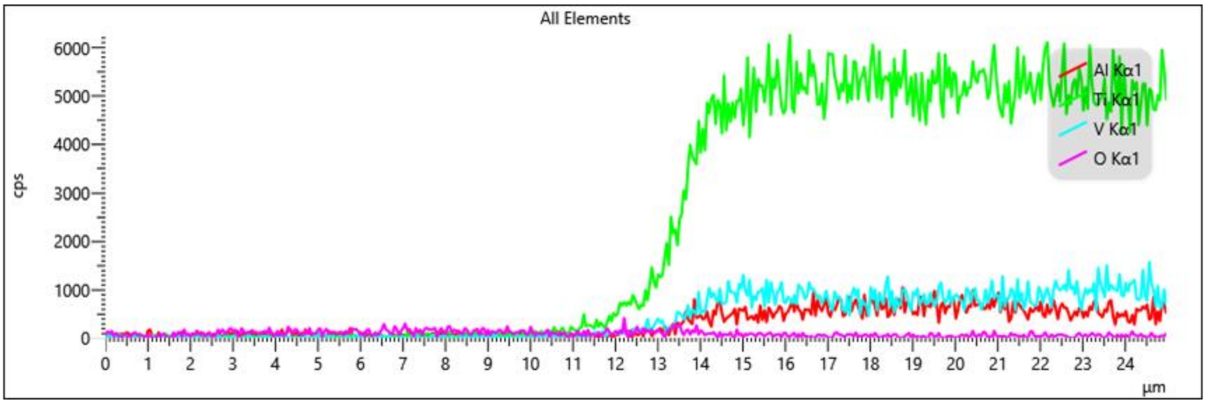
```matlab
figure(4)
plot(t_downsampled, Fxlp_downsampled, 'k', 'linewidth', 1)
xlabel('Time(s)')
ylabel('Fx(N)')
set(gca, 'FontSize', 16)
xlim([0, 150])
ylim([-400 400])
saveas(gcf, 'Fxlp_downsampled.png')
Fa_downsampled_positive = Fa_downsampled(Fa_downsampled>50);
Fa_downsampled_average = mean(Fa_downsampled_positive);

figure(5)
plot(t_downsampled, Fylp_downsampled, 'g', 'linewidth', 1)
xlabel('Time(s)')
ylabel('Fy(N)')
set(gca, 'FontSize', 16)
xlim([0, 150])
ylim([-400 400])
saveas(gcf, 'Fylp_downsampled.png')

figure(6)
plot(t_downsampled, Fzlp_downsampled, 'r', 'linewidth', 1)
xlabel('Time(s)')
ylabel('Fz(N)')
set(gca, 'FontSize', 16)
xlim([0, 150])
ylim([-400 400])
saveas(gcf, 'Fzlp_downsampled.png')

figure(7)
plot(t_downsampled, Fa_downsampled, 'b', 'linewidth', 1)
xlabel('Time(s)')
ylabel('Fa(N)')
```

```matlab
saveas(gcf, 'Fa_downsampled.png')



%% Average force
feed = 184; %unit: mm/min

n1 = find(Fxlp>25,1);
n2 = n1+50/feed*60*samf;
n3 = n2+90/feed*60*samf;
FxAveMMC = mean(Fxlp(n2:n3));
FyAveMMC = mean(Fylp(n2:n3));
FaAveMMC = mean(Fa(n2:n3));
FzAveMMC = mean(Fzlp(n2:n3));

Fx_max = max(Fxlp)
Fy_max = max(Fylp)
Fa_max = max(Fa);
Fz_max = max(Fzlp);

Fx_min = min(Fxlp)
Fy_min = min(Fylp)
Fa_min = min(Fa);
Fz_min = min(Fzlp);

% Fmeanmax = table(FxAveMMC, FzAveMMC, Fx_max, Fz_max,'VariableNames', { 'FxAve','FzAve
Fmeanmax = table(FaAveMMC, FzAveMMC, Fa_max, Fz_max,'VariableNames', { 'FaAve','FzAve',
writetable(Fmeanmax, 'ForceAverageAndMax.txt');
%[FxlpPKS, FxlpLOCS] =findpeaks (Fxlp);
```

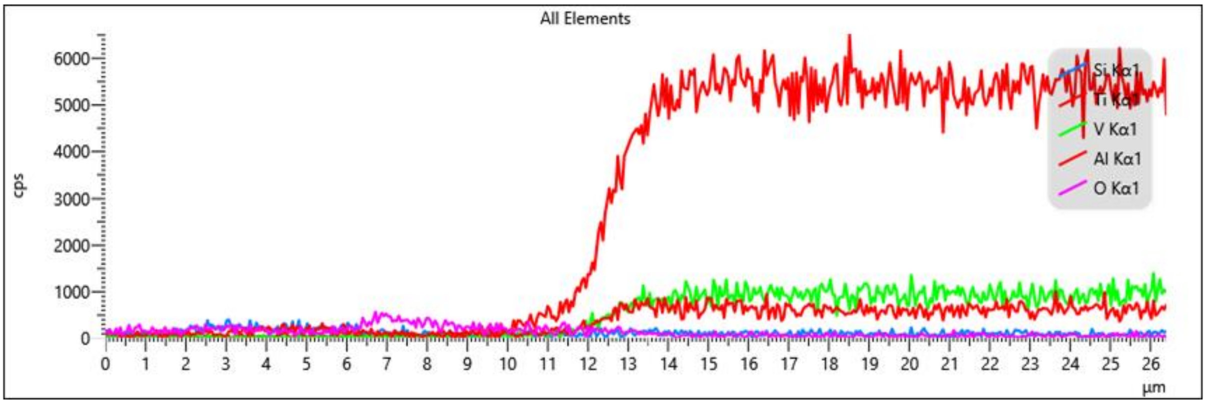
Figure 39. Matlab Code for cutting force analysis.

[redacted]

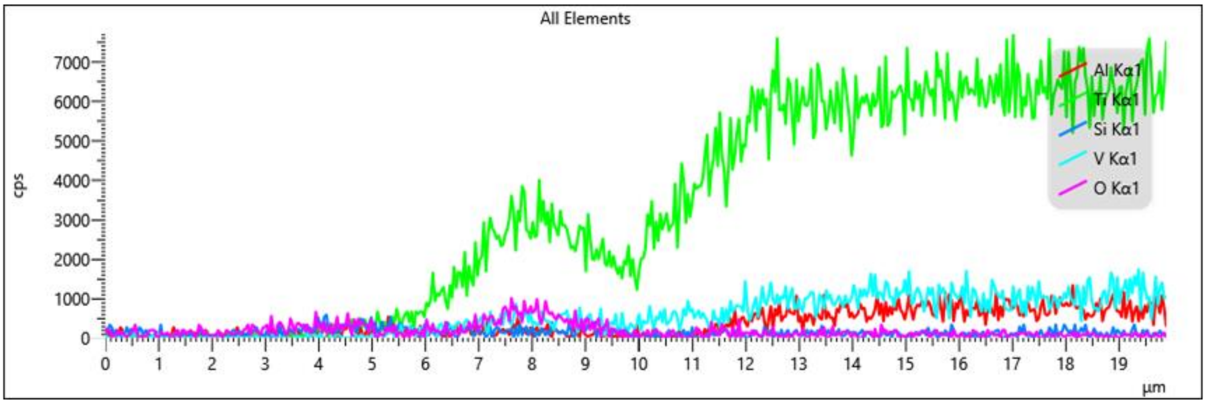
More results of EDX:



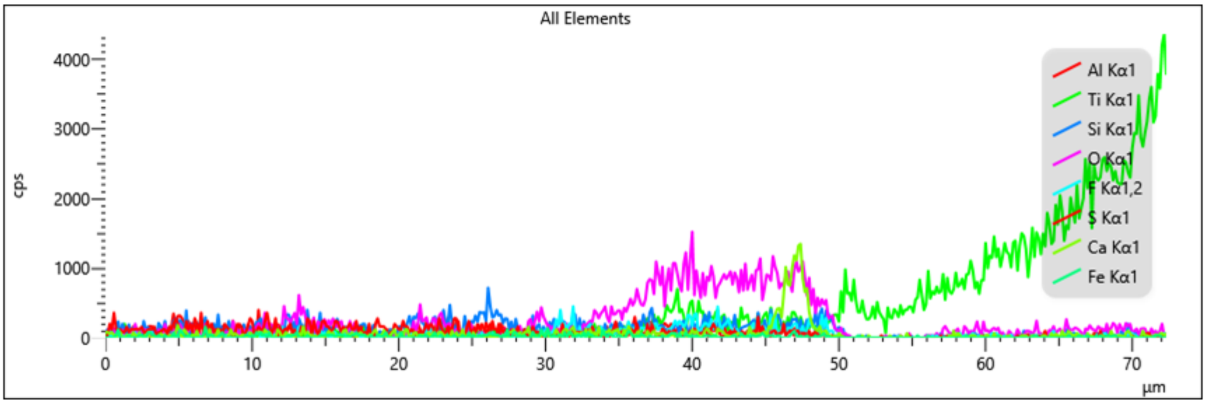
(a)



(b)



(c)



(d)

Figure 40. EDX inspection of the machined surface under a) 650 DOC LAMill; b) 1050 DOC LAMill; c) 650 DOC LS; d) 1050 DOC LS.

[redacted]